

Experimental investigation of parametric cell-to-cell variation and correlation based on 1100 commercial lithium-ion cells



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ABSTRACT

In order to prevent battery modules from inhomogeneity during operation, the integrated battery cells should be matched. Therefore, the cell-to-cell variations of relevant cell parameters due to manufacturing tolerances need to be quantified. Regarding cell matching, three points are lacking in current literature: first, reported parameter analyses have either based their statistic analysis on a too small number of cells. Or, second, few different cell parameters have been considered – hence, it was not possible to discuss which parameter would be beneficial to be used for classification. Third, potentially sensible correlations between different cell parameters have not been determined and discussed adequately. This paper provides a unique combination of analysing multiple different cell parameters and discussing their correlation – both based on a large number of examined cells: we investigate the parametric cell-to-cell variation and correlation of 1100 commercial production fresh LiFePO₄-graphite cells which originate from two batches. The cell parameters are experimentally determined by conducting DC check-up (CU)- and AC EIS-measurements under monitored temperature and relaxation conditions for all 1100 cells. The data is statistically analysed for 15 different parameters: different discharge capacities, different DC and AC impedances, the mass and mean temperature of the cells. The determined relative variation of capacity and impedance is small: 0.28% and 0.72% respectively, which corresponds to a variation ratio of 1:2.2. The variation of the cell impedance allows no conclusion about the variation of the cell capacity. From the results, we derive four major implications concerning recommended characterisation parameters for the development and modelling of battery modules as well as for the quality control during cell production. Our experimentally determined parametric variation values and drawn conclusions are valuable for model fittings and battery pack analyses which have up to now been based on assumptions about cell-to-cell variations. The data set for all 1100 cells and 15 parameters is provided as supplementary material [1].

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1. Introduction

Manufacturers of storage systems for electric cars and stationary storage applications aim to produce reliable and

long-lasting battery modules. In order to ensure these high quality standards, inhomogeneity within a module should be avoided [2]. Inhomogeneity in a battery module in this context denotes a gradient of a certain parameter across connected battery cells. There are different reasons which can lead to inhomogeneity within battery modules: the integration of unmatched cells showing cell-to-cell variations in capacity and impedance [2–7], uneven [8] or defective cell contacts [9,10], an ineffective cooling strategy [11] or an external heat source next to the battery module [12,13].

This work will investigate cell-to-cell variations of new cells. These can, in spite of high quality standards for commercial lithium-ion cells, result from manufacturing tolerances [3,14] and varying shipping or storage conditions. When integrating unmatched cells into battery modules, these variations can lead to

Abbreviations: AC, alternating current; CC, constant-current; CCCV, constant-current-constant-voltage; COC, coefficient of correlation; COD, coefficient of determination; COV, coefficient of variation; CTS, cell test system; CU, check-up; CV, constant-voltage; DC, direct current; DCH, discharge; ECM, equivalent circuit model; EIS, electrochemical impedance spectroscopy; fpd, frequencies per decade; LFP, lithium iron phosphate; mpf, measures per frequency; MU, measurement uncertainty; NCA, lithium nickel cobalt aluminium oxide; NMC, lithium cobalt nickel manganese oxide; OC, open circuit; SOC, state of charge; SOH, state of health.

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local temperature peaks and may result in inhomogeneous [3] and accelerated ageing [4,5]. Hence, parametric variations are crucial to consider when designing and assembling battery modules.

When aiming to determine the parametric variation of lithium-ion cells due to manufacturing tolerances, it must be ensured that possible cell-to-cell variations do not result from external reasons. Such reasons could be that the cells originate from different production batches, that the cells have been shipped under different conditions, thus experiencing a different calendar ageing, or that the cells have already experienced an operational history, i.e. have been cycled prior to the cell characterisation measurements, maybe even under different measurement conditions. Concluding, in order to being able to rely on experimentally determined parametric variations due to manufacturing tolerances, the cells to be used for investigation should be production fresh.

1.1. State of the art

The experimental determination of commercial cell-to-cell parametric variations has been addressed by a handful of research groups. However, measurements to determine cell-to-cell variations are time consuming: in order to allow for a thorough statistic analysis a large amount of cells has to be investigated. In terms of cell characterisation, this requires excessive measurement capacities.

Most publications, which base their work on a statistic analysis of battery cells, aim to investigate ageing behaviour [4,5,14–17] or the assembly and performance of battery modules [4–6,16–20]. In these cases the statistic analysis is mostly addressed marginally alongside the main focus of the respective work.

1.1.1. Number of investigated cells

Many publications base their statistic analysis on no more than 250 cells [2,5,14–16,18–21].

Dubarry et al. [2] have investigated 100 cells and measured their respective distribution of weight, initial (SOC) (as delivered), and discharge capacity. By simulation they identified cell internal attributes which resulted in the experimentally determined cell-to-cell variations: the amount of active material, polarisation resistance, and localised kinetic factors. Following this, they further investigated 10 cells towards their cell-to-cell variations [21] and chose a nominal sample cell in order to examine ageing phenomena in the cells [15].

Zheng et al. [19] have examined 96 cells regarding the variation of impedances in order to distinguish between the internal and contact resistance increase faults when the cells are connected in series. After having measured the variation of capacity of 16 cells, they connected 8 of them in series to a small battery module and parameterized a model which describes both, the battery module's state and the single cells' inconsistencies [20].

Miyatake et al. [18] have experimentally examined the influence of cell-to-cell variations in capacity on the discharge capacity of differently connected battery module topologies assembled from up to 12 cells: when cells are connected in series, the module capacity is affected by the cell with the smallest capacity. When cells are connected in parallel, the capacities of the single cells add up to the module capacity. When cells are connected in series and parallel, the module capacity depends on the module configuration.

Gogoana et al. [5] have measured the impedance variation of 72 cells in order to show that when two cells, which have a 20% different impedance, are connected in parallel, they experience an up to 40% higher ageing than two parallel-connected cells which have a similar internal resistance.

Campestrini et al. [16] compared battery module ageing to that of single cell ageing. For this they examined two battery modules consisting of 112 cells each which had been matched from a lot of 250 cells. Disassembling the battery modules allowed for an ageing analysis of capacity fade and impedance rise regarding local temperature distribution and cell-to-cell variation within the modules.

Baumhöfer et al. [14] have analysed the variation of capacity (constant-current-constant-voltage (CCCV)) of 48 commercial cells at two states of ageing. Moreover, they determined different pulse and alternating current (AC) impedances in order to identify parameters to be measured at the begin-of-life in order to predict ageing behaviour. They found that the deviations between the capacities of the aged cells increase compared to the capacity distribution of the new cells. They point out that using more cells for the statistic investigation, would be beneficial [14].

Some research groups do examine larger lots of cells [4,6,17,22,23].

Paul et al. [4] have investigated 20,000 lithium-ion cells. They measured the initial cell capacity and the direct current resistance. Using 96 of these cells they furthermore examined ageing inhomogeneity in battery systems due to both, cell-to-cell variations and temperature gradients and validated a battery module simulation approach based on the Monte Carlo method.

An et al. [22] have measured the rate dependency of cell-to-cell variations of lithium-ion cells. Therefore they analysed the distribution of cell capacity and mass based on 5473 cells. From these they used 198 cells to additionally examine the direct current (DC) impedance and correlate both parameters to the current rate. From the capacity distribution of 7739 cells they moreover matched cells to build parallel-connected battery modules. Using these, they validated their ECM and studied the effect of cell sorting methods on the battery modules performance [6].

Schuster et al. [17] have analysed 484 new and 1908 aged lithium-ion cells from two identical battery electric vehicles towards their capacity and impedance distribution: the distribution shapes change during ageing from normal towards Weibull distribution. Following this, having complemented the data set by additional 83 partly aged cells, they furthermore examined the correlation between capacity and impedance of lithium-ion cells during calendar and cycle life in order to conclude upon whether testing the cell impedance can be used to deduce the cell capacity [23]: the correlation behaviour is strongly dependent on operational and storage conditions.

1.1.2. Number of investigated parameters

What becomes clear: most papers do only examine the variation of one DC or AC impedance and one discharge capacity – sometimes even without clearly specifying whether the discharge capacity has been determined during the constant-current (CC) phase or the CCCV phase. Only a small number of publications determine the cell-to-cell variations of different impedances [14,17]; none of them reports on a comparison of different capacities. However, such a comparison for both, capacities and impedances, seems crucial to us in order to make sure that the determined parameter during cell characterisation is the most promising to be used in order to reveal the desired information. What is more, discussing the dependency between different external cell parameters, i.e. a certain cell capacity and impedance, can help to reveal dependent parameters of which one could in consequence be omitted during cell characterisation and which would thus save measurement time.

1.1.3. Correlation between parameters

The dependency between different external cell parameters can be depicted by investigating the correlation between these

parameters. Schuster et al. [23] have shown the correlation between two different AC impedances and the discharge capacity respectively. They compared the correlation plots determined from new cells to those of differently aged additional cells. They have neither correlated the two AC impedances to each other nor have they regarded DC impedances, different discharge capacities or other external cell parameters. Schweiger et al. [24] have determined the internal resistance of a cell by using different DC and AC methods and discussed the results qualitatively. However, as only one single cell was tested, they have not been able to show correlation graphs of the different impedances which would have been beneficial in order to quantify the dependency between the investigated parameters. Baumhöfer et al. [14] have examined the respective real and imaginary part as well as the frequency of an AC impedance and an additionally determined pulse resistance at the begin of life. Subsequently they correlated these parameters to the calculated lifetime of their investigated number of cells. However, they have not correlated the parameters themselves. Hence, an extensive study of the correlation between multiple different external cell parameters of new cells has not yet been reported.

By simulation, correlations between cell parameters have already been addressed: by developing an impedance model, Santhanagopalan et al. [25] have investigated the dependency of the variability of certain internal cell parameters (such as electrode thickness, porosity, tortuosity and particle size) on the parametric variation of the AC impedance and thus on the performance at cell level. Kenney et al. [26] have simulated capacity variations in electrode manufacturing by varying the electrode thickness, electrode density and active material weight fraction and discussed the impact of these on the resulting capacity of serial-connected lithium-ion cells. However, they have not correlated the parameters.

The variation of cell internal parameters is the reason for the variation of cell external parameters due to manufacturing tolerances. However, as the experimental determination of these cell internal parameters is difficult, it is not suitable for cell characterisation on a big scale. Instead, determining the cell external parameters for the purpose of cell sorting and matching during battery module assembly is beneficial.

1.2. Originality and aim of this paper

In order to investigate parametric cell-to-cell variation and correlation between external cell parameters, using and comparing cell-to-cell variation data from different literature sources is not sensible. Rather should the measurements, parameter determinations and data analysis be from one literature source.

Indeed, this is the originality of our work: we provide results regarding (1) cell-to-cell variations of multiple external cell parameters such as cell capacities, impedances, mass and temperature, (2) and examine the dependency between these parameters by discussing their correlation, (3) while meeting the requirements of a thorough statistic analysis by basing our investigation on a sufficient number of cells, (4) and strictly monitoring the measurement conditions including the temperature as well as the relaxation times between the different characterisation measurements for each single cell. (5) From the analysis we extract implications for the development and modelling of battery modules and for the quality control during cell production.

More specific, the aim of this paper is to experimentally investigate the cell-to-cell variation of different parameters and the correlation between them. In order to allow for a valid statistic analysis, 1100 commercial 26650 LiFePO₄-graphite cells are examined, which originate from two batches. Hence, the influence

of different batches on the parametric variation and correlation can additionally be investigated. We conduct both, DC and AC measurements and we strictly monitor the measurement conditions including the temperature as well as the relaxation times between the different characterisation measurements for each single cell. We determine multiple cell parameters: different discharge capacities, different DC and AC impedances, the mass and the temperature of the cells. The determination of these different parameters allows for a comparison of their variations and to find correlations between the parameters. Thus, for example, we are able to compare the variation of discharge capacity to the variation of impedance of a cell, moreover correlate these parameters and discuss the result in context of the cell temperature. We furthermore determine the measurement uncertainties for the parameters and discuss their influence on the variation results. We provide distinct quantitative statements about the parametric variation of and the correlation between the investigated parameters and use these results to extract implications for the development and modelling of battery modules and for the quality control during cell production.

1.3. Scopes of application

Five main scopes of application can be identified of our work: (1) cell matching for the design and assembly of battery modules, (2) modelling battery modules, (3) designing balancing systems and monitoring methods for battery modules, (4) designing cooling strategies for battery modules and (5) manufacturing process quality control.

1.3.1. Cell matching for the design and assembly of battery modules

For the design and assembly of battery modules, cell capacity and impedance are crucial parameters which should be investigated regarding cell-to-cell parametric variations: when connecting battery cells in series, the capacity of the weakest cell specifies the capacity of the battery system [18,27] and moreover its lifetime [28]. When connecting cells in parallel, a variation in the cell impedances results in different currents through the cells [9,29,30], resulting in different states of charge (SOCs), temperatures and degradation rates during operation although their terminal voltage is at the same level [3]. A prediction about the development of cell-to-cell variations within a battery module due to operation and cell ageing is not possible only based on the knowledge about initial cell-to-cell variations of fresh cells. Yet, the use of similar cells during the assembly of battery modules is still advantageous and recommended: similar cell capacities allow for a best possible utilisation of available capacity and similar cell impedances form the basis for evenly distributed currents within a module from the beginning of module operation. Hence, cell matching during the assembly of battery modules is crucial in order to minimize inhomogeneity in battery modules due to cell-to-cell variations [31].

Of course, the cell-to-cell variations need to be determined for the respective cells which are to be integrated into the battery module. Still, our work allows a conclusion about which parameters are suitable for cell matching during the assembly of battery modules. Hence, only the most crucial parameters need to be determined during cell characterisation measurements and thus measuring time and resources can be saved.

1.3.2. Modelling battery modules

Another scope of application for a set of cell-to-cell variation data is to integrate it into models for battery modules and packs. This will help to enhance the prediction accuracy of the model [21]. In absence of a profound data set concerning parametric cell-to-cell variation due to manufacturing tolerances, cell parameter

variations have often been chosen randomly within a reasonable range in order to being able to investigate the influence of cell-to-cell variations on the performance of connected cells or battery modules by simulation [26].

What is more, a detailed statistics study regarding the dependency between one parameter and another, e.g. cell impedance and temperature, has been identified as beneficial for the application of developed experimental and simulation methods [27].

Our work can be used as a data basis when modelling connected cells and battery modules.

1.3.3. Designing balancing systems and monitoring methods for battery modules

Cell-to-cell variations affect the performance in battery modules [32]: the inhomogeneity in battery modules which arises from cell-to-cell variations due to manufacturing tolerances, evoke the need of cell equalization in battery modules [33], the parameters of the single cells or cell strings need to be estimated [32]. Although balancing systems mostly estimate the SOC of the cells [34,35], Wang et al. [32] point out, that inhomogeneity within battery packs should be equalized not only concerning cell voltage but also pack voltage, temperature, maintenance current and impedance. A temperature difference between parallel-connected cells results in unbalanced discharging and ageing of the cells [36] and affects the battery module's performance [12,37]. Concerning ageing, capacity and impedance data of non-uniform cells is also needed for demonstrating (SOH) monitoring methods on battery modules when aiming to investigate the effects of cell non-uniformity [38].

During the design of balancing systems and SOH monitoring methods for battery modules, our data set can serve as input for the model: the distinct quantitative statements about the parametric variation of different cell external parameters provide an experimentally determined range of parametric variation and cell non-uniformity.

1.3.4. Designing cooling strategies for battery modules

Cell-to-cell variation can lead to inhomogeneous operation of the cells within battery modules, which increases the temperature gradient across the connected cells and might even lead to the failure of a cell. This gives rise to investigations on an optimised module design and cooling strategy [39]. For optimising a battery pack towards mass, volume and material cost, considering cell-to-cell variations in the model might not be of large importance [40]. However, regarding the optimisation of a battery module's performance by improving module design and cooling strategies [11], considering cell-to-cell variations and thereby arising temperature gradients across the module (additionally to those arising from the module geometry), would be beneficial [39].

Our work can serve as data base when designing cooling strategies for battery modules.

1.3.5. Manufacturing process quality control

Cell-to-cell variation in cell external parameters, which is due to manufacturing tolerances, results from a variation in the cell internal parameters [2]. However, both, cell capacity and impedance are influenced by multiple cell internal parameters. Our work can be used to investigate the relation between the variation of cell internal and external parameters by simulation, conclude on the main influencing factors on capacity and impedance and thus aid to improve the manufacturing process quality control.

In order to improve the manufacturing process of one specific lithium-ion cell, experimental data of the respective cell is needed. However, the analysis of our data set and drawn implications will

help to decide upon which parameters are sensible to be determined during cell characterisation.

1.4. Structure of this paper

The paper is structured as follows: first, the statistic theory is presented, which is used for the statistic analysis of the cell parameters. This is followed by describing the experimental settings including the measurement equipment with its corresponding measurement precision, the relaxation and thermal conditions, the settings of the DC and AC measurements, possible sources of error, the definition of the investigated parameters and their measurement uncertainties. The results are illustrated, described and discussed in three parts: first, the parametric variations of the cell parameters is presented. Second, the parametric correlations between the cell parameters are analysed. Third, implications for the quality control during cell production as well as for developing and modelling battery modules are derived. Finally, the main findings are concluded and the impact of the presented work is highlighted. The data set for all cells and parameters is provided as supplementary material [1].

2. Statistic theory

For the statistic analysis of the measured cell parameters, statistic theory for parametric variation and correlation is used based on [41].

2.1. Parametric variation

It is assumed that all cell parameters are normally distributed around their respective mean value μ [4,16,17].

The mean value of a parameter x calculates

$$\mu_x = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

where n is the number of data points, i.e. the number of investigated cells. In a normal distribution the parameter's standard deviation σ around the mean is defined as

$$\sigma_x = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \mu_x)^2} \quad (2)$$

68.1% of the values lie within the interval $\pm\sigma_x$ around μ_x .

The coefficient of variation (COV) σ_x/μ_x is a relative measure of a parameter's variation independent of scale (relative variation). It is thus used to compare the variation of different parameters in order to set them in context to each other.

2.2. Parametric correlation

Investigating the correlation between two parameters using a set of n cells offers a method to describe the dependency between the two parameters. The correlation coefficient ρ_{xy} of two variables x and y calculates

$$\rho_{xy} = \frac{\sum_{i=1}^n (x_i - \mu_x)(y_i - \mu_y)}{\sqrt{\sum_{i=1}^n (x_i - \mu_x)^2 \cdot \sum_{i=1}^n (y_i - \mu_y)^2}} \quad (3)$$

with μ_x and μ_y being the mean values of the two parameters. It measures the strength of the linear dependency of two parameters. The correlation coefficient ranges from -1 to 1 . $\rho_{xy} = 0$ means that there is no correlation between parameter x and y , $\rho_{xy} = 1$ implies a total positive and $\rho_{xy} = -1$ a total negative linear dependency respectively. In addition, correlation coefficients are in this paper

classified as follows:

$$\begin{aligned} |\rho_{xy}| \leq 0.3 & \quad \text{weak correlation} \\ 0.3 < |\rho_{xy}| \leq 0.7 & \quad \text{moderate correlation} \\ 0.7 < |\rho_{xy}| & \quad \text{strong correlation} \end{aligned} \quad (4)$$

These classes have been chosen by the authors due to the coefficient of determination (COD) of the class limits. The COD ρ_{xy}^2 denotes the ratio of dependency of the variation of parameter x on the variation of parameter y . For a correlation coefficient $\rho_{xy} = 0.3$ between two parameters, the COD equals $\rho_{xy}^2 = 0.09 = 9\%$. Thus a weak correlation between two parameters x and y means that up to 9% of the variation of parameter x depends on the variation of parameter y . For $\rho_{xy} = 0.7$ the COD equals $\rho_{xy}^2 = 0.49 = 49\%$. Thus a strong correlation means that more than 49% of the variation of parameter x depends on the variation of parameter y . All correlation values in between $0.3 < |\rho_{xy}| \leq 0.7$ are classified as a moderate correlation.

3. Experimental

In order to investigate the distribution of several cell parameters and the correlation between them, check-up (CU)- and electrochemical impedance (EIS)-measurements have been performed under monitored conditions.

3.1. Investigated cell

The cell under investigation is a commercial 3 Ah 26650 LiFePO₄-graphite cell (SonyUS26650FTC1). 1100 fresh cells of two batches (batch B1: cell 1–600, batch B2: cell 601–1100) have been examined. The batches are defined by a code printed on each individual cell and which corresponds to the production date. At the time of investigation the cells of batch B1 had an age of 9 months, the cells of batch B2 of 10 months. Table 1 shows the nominal data of the investigated cell.

3.2. Measurement equipment

The check-up (CU)-measurements have been performed using a BaSyTec cell test system (CTS) and an Espec LU-123 climate chamber at $T = 25^\circ\text{C}$. The surface temperature of each cell has been monitored using PT100 thermocouples. For the electrochemical impedance spectroscopy (EIS)-measurements we used a BioLogic VMP3 potentiostat and a climate chamber at $T = 25^\circ\text{C}$. The weight measurements have been executed using a precision balance Sartorius Talent TE313S at room temperature. The precision of the measurement equipment is stated in Table 2.

During both, the CU- and the EIS-measurement, the cells have been contacted by custom-made terminal boards using gold plated spring contacts and enabling a four-wire-measurement.

Table 1
Nominal data of the investigated cell Sony US26650FTC1.

Parameter	Symbol	Value
Nominal capacity	C_N	3 Ah
Nominal voltage	V_N	3.2 V
Max. charge voltage	$V_{ch,max}$	3.60 ± 0.05 V
Min. discharge voltage	$V_{disch,min}$	2 V
Ohmic impedance (1 kHz, AC)	$R_{i,AC}$	18 m Ω
Max. continuous charge current	$I_{ch,max}$	2.85 A
Max. continuous discharge current	$I_{disch,max}$	20 A
Temperature range charge	$T_{amb,ch}$	0°C to 45°C
Temperature range discharge	$T_{amb,disch}$	-20°C to 60°C
Mass	m_{cell}	84.5 g

Table 2

Precision of the measurement equipment. During the CU-measurement the sampling rate was 0.1 Hz.

Instrument	Variable	Precision	Range
BasyTec CTS	Time	$\Delta t = \pm 1 \mu\text{s}$	
	Voltage	$\Delta V = \pm 0.3 \text{ mV}$	$V \in [0 \text{ V}, 6 \text{ V}]$
	Current	$\Delta I = \pm 200 \mu\text{A}$	$I \in [300 \text{ mA}, 5 \text{ A}]$
		$\Delta I = \pm 10 \mu\text{A}$	$I \in [15 \text{ mA}, 300 \text{ mA}]$
		$\Delta I = \pm 0.5 \mu\text{A}$	$I \in [1 \text{ mA}, 15 \text{ mA}]$
BioLogic VMP3	Phase angle	$\Delta \phi = \pm 1^\circ$	$f \in [10 \mu\text{Hz}, 1 \text{ MHz}]$
	Impedance	$\Delta Z / Z = \pm 1\%$	$f \in [10 \mu\text{Hz}, 1 \text{ MHz}]$
	Temperature	$\Delta T (@25^\circ\text{C}) = \pm 0.2 \text{ K}$	$T \in [-70^\circ\text{C}, 250^\circ\text{C}]$
Sartorius Talent	Mass	$\Delta m = \pm 0.001 \text{ g}$	$m \in [0 \text{ g}, 310 \text{ g}]$

3.3. Relaxation and thermal conditions

During the measurements special care has been taken of each individual cell concerning relaxation times between CU- and EIS-measurements and their storage temperature: from the time of arrival at the institute the cells had been stored at $T \approx 8^\circ\text{C}$. At least 3 h before the start of the CU-measurement, the respective cells have been stored at room temperature ($T \approx 25^\circ\text{C}$). 64 cells have been tested at a time. After the CU-measurement had been finished, the cells were stored at room temperature again for a minimum of $t = 40$ – 48 h. After this waiting-time between CU- and EIS-measurements, cell internal relaxation processes should have declined sufficiently [42]. Directly before the EIS-measurement was started, the cells have been stored at $T = 25^\circ\text{C}$ in a climate chamber for at least 3 h. An overview of the measurement and storage sequence illustrating the temperature condition and time duration for each step is presented in Fig. 1.

3.4. CU-measurement

The cells have been shipped to the institute at a SOC $\approx 30\%$. First, each cell has been weighed as delivered (including its shrink sleeving). The CU-measurement first charged the respective cell using a CCCV protocol to 3.6 V using a CC $I_{ch,CC} = 1 \text{ C} = 3 \text{ A}$ followed by a constant-voltage (CV) (until $|I_{ch,CV}| < C/30 = 100 \text{ mA}$). This was followed by a 1 min 1 C discharge pulse and a 6 min open circuit (OC) period. Afterwards the cell has again been CCCV-charged. This charge period has again been followed by a 6 min OC. Then, the cell has been CCCV-discharged to 2 V using a CC $I_{disch,CC} = 1 \text{ C} = 3 \text{ A}$ (CV until $|I_{disch,CV}| < C/20 = 150 \text{ mA}$). During these discharge (DCH) periods (CC and CV) the discharge capacities of the cell $C_{DCH,CC}$, $C_{DCH,CV}$ and $C_{DCH,CCCV}$ are determined by Ah-counting. Having allowed for a short relaxation period of 6 min OC, the cell is finally CC charged to SOC = 50% using $I_{ch,CC} = 1 \text{ C} = 3 \text{ A}$, which corresponds to $0.5 C_{DCH,CCCV}$. The CU-plan is concluded by another 6 min relaxation period at OC. The relevant part of the test protocol, from which the parameters are defined, is displayed in Fig. 2 left.

3.5. EIS-measurement

The EIS-measurement investigates the cell impedance within a logarithmic spaced frequency range $f \in [10,000, 0.01] \text{ Hz}$ using an excitation current amplitude of $\hat{I} = 50 \text{ mA}$. The number of frequencies per decade (fpd) and number of measures per frequency (mpf) changes according to the chosen sub-intervals: $f \in [10,000, 1] \text{ Hz}$ (13 fpd, 10 mpf), $f \in [1, 0.1] \text{ Hz}$ (10 fpd, 5 mpf), $f \in [0.1, 0.01] \text{ Hz}$ (3 fpd, 2 mpf). The test is performed at SOC = 50% and $T = 25^\circ\text{C}$. The data points at high frequencies representing an inductive behaviour are neglected during the analysis (comp. Fig. 2 right).

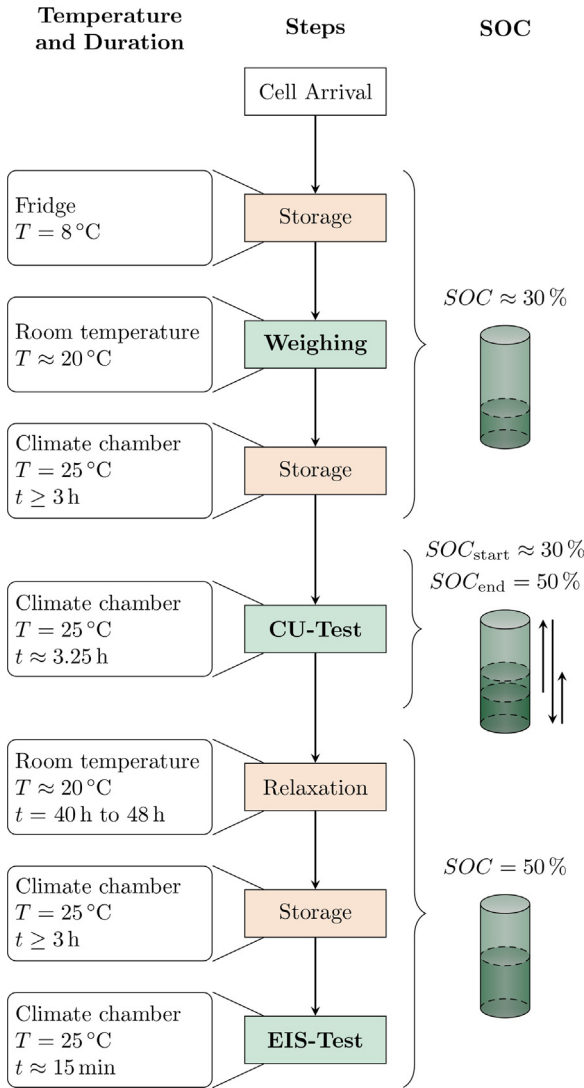


Fig. 1. Flow chart of the measurement and storage sequence illustrating the temperature condition and time duration of each step.

3.6. Possible sources of errors

There are different sources of error which could externally influence the variation of parameters and should therefore be kept in mind during the interpretation of results. Possible observations within the parametric variation over the cell number could be group building, drifts or trends.

The cells have been shipped in boxes of 25 cells. We numbered the boxes according to their position in the delivered package and we numbered the cells according to their position in the respective box. Hence, if the variation plot of a certain parameter shows groups of 25 cells within a certain parameter range, this could be due to the packaging order of the boxes.

The cells have been measured in sequential order. For the CU-measurement 64 cells have been measured at a time. For the EIS-measurement 8 cells have been measured at a time. The position of the cell regarding the terminal board, measurement channel, and position in the climate chamber could all result in a specific temperature influence which differs from the other cells. An increased cell temperature would for example result in an increased discharge capacity. Therefore, the parametric variation of the temperature over cell number should be analysed carefully

regarding drifts or trends. The correlation coefficient of other parameters with the temperature can reveal possible dependencies.

Measurement uncertainties (MU) influencing the determined parameters will be discussed in Chapter 3.8.

3.7. Definition of parameters

In total, 68 parameters have been evaluated using both, CU- and EIS-measurements. In this paper, we focus on 15 of those parameters: mass and mean temperature during measurement of the cell, three different discharge capacities, seven different impedances obtained from both, CU- and EIS-measurement and the interpolated frequencies of the EIS impedances. Fig. 2 shows representative CU- and EIS-measurement data of one cell (left: last part of the CU-measurement, right: Nyquist diagram of the EIS-measurement) and defines the obtained capacity and impedance parameters, which are discussed in this paper.

Three values of discharge capacity are investigated. They are obtained from the last part of the CU-measurement starting at the begin of discharge: $C_{\text{DCH_CC/Ah}}$ is the capacity obtained during the discharge CC-phase, $C_{\text{DCH_CV/Ah}}$ is the capacity obtained during the discharge CV-phase and $C_{\text{DCH_CCCV/Ah}}$ is the capacity obtained during both, the discharge CC- and CV-phase. A comparison of the three values will allow to analyse the influence of the variation of cell impedance on the variation of cell capacity.

Different values of impedance have been obtained from the two measurements: the DC impedances $R_{\text{DC_10s}}$, $R_{\text{DC_30s}}$ and $R_{\text{DC_60s}}$ are defined as the ratio between the difference in voltage and the difference in relaxation current at the end of the CU-measurement. The values are obtained after the final constant current charge phase. The time interval is 10 s, 30 s and 60 s respectively. After $t = 10\text{ s}$ influences of the ohmic drop, charge transfer and diffusion processes contribute to the DC impedance, after $t = 30\text{ s}$ and $t = 60\text{ s}$ the influence of diffusion has become more dominant. Due to this fact, the SOC and temperature sensitivity of the DC impedance increases for $R_{\text{DC_30s}}$ and $R_{\text{DC_60s}}$ compared to $R_{\text{DC_10s}}$. Comparing the variation results of these three parameters, the change of DC impedance variation over time can be investigated. Comparing their correlation results, the changing temperature sensitivity can be depicted.

The AC impedances are defined within the Nyquist diagram and are determined by interpolating the measured EIS-data to find the characteristic points $R_{\text{AC_Im0}}$, $R_{\text{AC_1000Hz}}$, $R_{\text{AC_max}}$ and $R_{\text{AC_min}}$. $R_{\text{AC_Im0}}$ is defined as the zero crossing of the data with the real axis: $R_{\text{AC_Im0}} = \text{Re}(Z)|_{\text{Im}(Z)=0}$ at an interpolated frequency $f_{\text{AC_Im0}}$. $R_{\text{AC_1000Hz}}$ is defined as the real part of the impedance at an interpolated frequency $f = 1000\text{ Hz}$: $R_{\text{AC_1000Hz}} = \text{Re}(Z)|_{f=1000\text{ Hz}}$. $R_{\text{AC_max}}$ is defined as the real part of the impedance between $R_{\text{AC_Im0}}$ and the local maximum at an interpolated frequency $f_{\text{AC_max}}$. $R_{\text{AC_min}}$ is defined as the real part of the impedance between $R_{\text{AC_Im0}}$ and the local minimum at an interpolated frequency $f_{\text{AC_min}}$. The $R_{\text{AC_max}}$ and $R_{\text{AC_min}}$ impedances are chosen for investigation as they are frequently used for fitting purposes of equivalent circuit models (ECM). The $R_{\text{AC_1000Hz}}$ impedance is analysed, as manufacturers often measure this impedance in order to determine the ohmic impedance of a cell. This is due to the fact that measurement devices can more easily detect the impedance at a constant frequency (i.e. $f = 1000\text{ Hz}$) than to find the true ohmic impedance $R_{\text{AC_Im0}}$. The latter is more often considered in scientific publications. This work will allow for a comparison of both ohmic AC impedance values.

In addition to the capacity and impedance values, the mass and temperature of the cells are considered in the data analysis. T_{Mean} is the mean surface temperature of the cell during the CU-

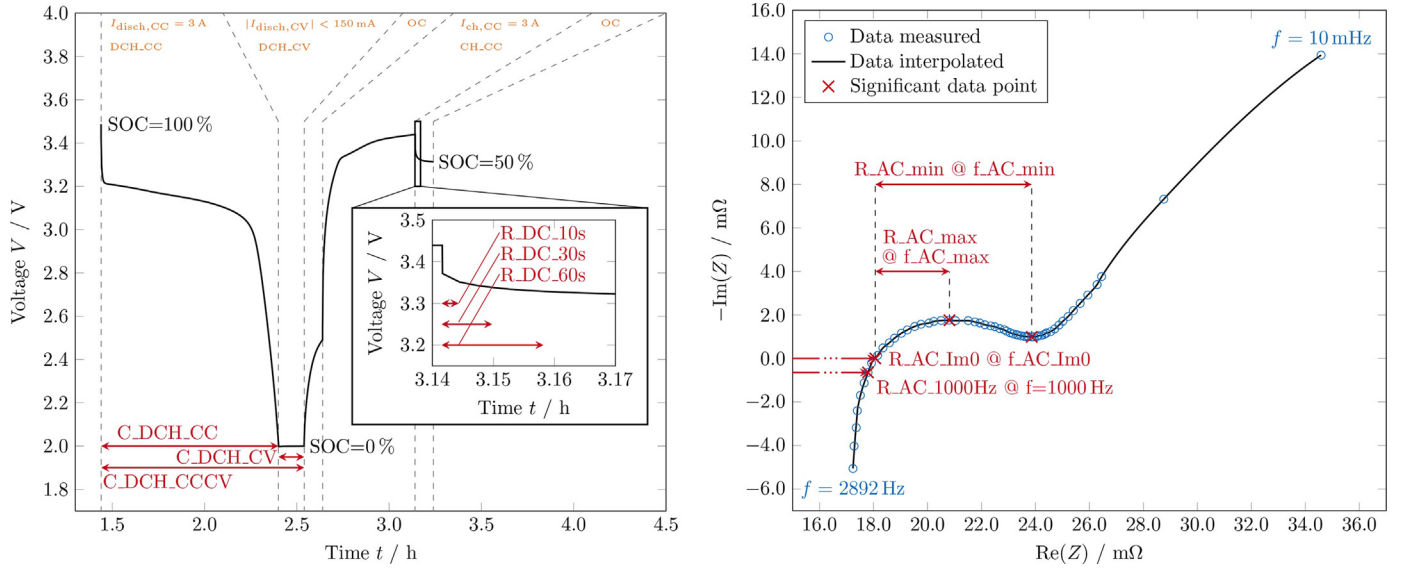


Fig. 2. Definition of parameters from the last part of the CU-measurement (left) and Nyquist diagram of the EIS-measurement (right). Note, that the imaginary axis of the Nyquist diagram is displayed reversed. Hence, the so called local maximum impedance is in fact a minimum and vice versa. Additionally the mass and temperature of each cell is analysed.

measurement. The variation of T_{Mean} is used to discuss the possible error sources of cell positioning within the climate chamber, the choice of measurement channel or terminal board during the CU-measurement. Correlation between T_{Mean} and another parameter is only to some extent sensible when the second parameter has not been determined from the CU-measurement.

3.8. Measurement uncertainties

In order to being able to discuss the statistic values of a parameter, these values must be related to the measurement uncertainty (MU) of the parameter. Only the mass and the temperature are determined directly. All other investigated parameters are calculated from measured basic quantities (i.e. time, current, voltage). In this case, the measurement uncertainty (MU) of a parameter x depends on the MU of the measured basic quantities. If a parameter x calculates from a product or quotient of these basic quantities $x = a \cdot b$, then the corresponding MU $u(x)$ calculates [43]:

$$u(x) = \sqrt{\left(\frac{\partial x}{\partial a}\right)^2 u^2(a) + \left(\frac{\partial x}{\partial b}\right)^2 u^2(b)} \quad (5)$$

where $\frac{\partial x}{\partial a}$ and $\frac{\partial x}{\partial b}$ are the partial derivatives of x with respect to a and b respectively and $u(a)$ and $u(b)$ are the MU of the basic quantities.

If a parameter x calculates from a sum or difference of these basic quantities $x = a + b$, then $u(x)$ calculates from the sum of the absolute errors in the quantities a and b :

$$u(x) = u(a) + u(b) \quad (6)$$

The MU of a basic quantity is determined based on the given MU range stated in the data sheet. If only the lower and upper limits of the MU of a directly measured quantity $u(a) \in [m, n]$ are given and if moreover no information is given about the distribution of the MU, then a uniform distribution is to be assumed between the limits [43]. In this case the variance of the corresponding MU $u^2(a)$ is given by [43,44]:

$$u^2(a) = \frac{(n - m)^2}{12} \quad (7)$$

It is assumed, that the MU of a parameter x is the same for each single determination of the respective parameter for all cells. Hence, the MU is also the same when relating it to the mean or standard deviation of the parameter. The relations $u(x)/\mu_x$ and $u(x)/\sigma_x$ allow for a discussion of the influence of the MU on the parametric variation result. Both values are given in % and are stated in the results table (comp. Table 3) for those parameters, for which the required MUs have been given in the data sheets. Note, that in order to not influence the determined parametric variation results, they are not corrected for the MU of the parameter. Instead, the respective MU values will be used to discuss the results.

4. Results and discussion

The obtained CU- and EIS data is evaluated and analysed in two steps: First, the distribution of the parameters is displayed and their variation is statistically analysed in Section 4.1. Second, the correlation between the investigated parameters is depicted and discussed in Section 4.2. The results of the parametric variations are summarized in Table 3, those of the parametric correlations are shown in Table 5.

4.1. Parametric variations of cell parameters

Regarding the parametric variation, each parameter is investigated based on its mean μ , standard deviation σ around the mean and its COV σ/μ . For each parameter a normal distribution is assumed [4]. The results of the parametric variations are displayed in different comparative figures (comp. Figs. 3–8) and are additionally summarized in Table 3.

All result figures showing the variation of a certain parameter based on one batch are structured as follows: the data is presented in two different graphs. The left graph shows the statistic variation of the parameter in a histogram plot, relating the number of cells to a certain parameter value. The right graph displays the respective parameter value over the sequential cell number (B1: 1–600, B2: 601–1100) in a scatter plot. In both graphs, the purple lines give the respective mean value μ , the green lines mark intervals, each with a width of the standard deviation σ of the displayed parameter. The values of mean μ , standard deviation σ around the mean and

Table 3

Experimentally determined parametric variations of the two batches. The variation is given in terms of mean value μ , standard deviation σ around the mean and coefficient of variation (COV) σ/μ . The measurement uncertainty (MU) of the parameter u is related to both, μ and σ of the parameter. The definition of the examined parameters is shown in Fig. 2. If no value is given, information to calculate the value have been missing.

Parameter x	Batch	μ_x	σ_x	σ_x/μ_x	$u(x)/\mu_x$	$u(x)/\sigma_x$
Mass and Temperature						
Mass/g	B1	84.671	0.063	0.07%	<0.01%	0.92%
	B2	84.716	0.098	0.12%	<0.01%	0.59%
T_Mean/°C	B1	27.440	0.702	2.56%	0.42%	16.45%
	B2	27.559	0.706	2.56%	0.42%	16.36%
Discharge capacity						
C_DCH_CC/Ah	B1	2.911	0.012	0.41%	<0.01%	0.93%
	B2	2.914	0.014	0.48%	<0.01%	0.80%
C_DCH_CV/Ah	B1	0.109	0.009	7.94%	0.01%	0.14%
	B2	0.104	0.009	8.65%	<0.01%	0.10%
C_DCH_CCCV/Ah	B1	3.020	0.007	0.23%	<0.01%	1.78%
	B2	3.018	0.010	0.33%	<0.01%	1.12%
DC Impedances (CU)						
R_DC_10s/mΩ	B1	28.485	0.407	1.43%	0.40%	28.34%
	B2	28.744	0.352	1.23%	0.40%	32.77%
R_DC_30s/mΩ	B1	32.861	0.438	1.33%	0.35%	26.33%
	B2	33.145	0.403	1.22%	0.35%	28.62%
R_DC_60s/mΩ	B1	35.805	0.469	1.31%	0.32%	24.59%
	B2	36.109	0.442	1.22%	0.32%	26.10%
AC Impedances (EIS)						
R_AC_1m0/mΩ	B1	17.642	0.320	1.82%	0.58%	32.55%
	B2	17.729	0.130	0.73%	0.57%	78.26%
R_AC_1000Hz/mΩ	B1	17.389	0.310	1.78%	0.59%	33.21%
	B2	17.464	0.123	0.70%	0.58%	81.86%
R_AC_max/mΩ	B1	2.799	0.222	7.93%	4.39%	55.36%
	B2	2.667	0.223	8.36%	4.39%	52.51%
R_AC_min/mΩ	B1	5.615	0.146	2.60%	2.46%	94.61%
	B2	5.374	0.165	3.08%	2.50%	81.26%
Frequencies						
f_AC_1m0/Hz	B1	768.612	35.082	4.56%	–	–
	B2	753.562	28.496	3.78%	–	–
f_AC_max/Hz	B1	90.438	13.934	15.41%	–	–
	B2	95.723	15.616	16.31%	–	–
f_AC_min/Hz	B1	3.323	0.689	20.73%	–	–
	B2	3.897	0.879	22.56%	–	–

coefficient of variation σ/μ are given in the respective upper left corner of each graph. The black curve in the histogram plot depicts the normal distribution curve.

The respective histogram plot can be used to depict group building within the batches. Due to the sequentially cell numbering, the scatter plot can be used to reveal a possible drift within the measurement, dependency on cell positioning or dependency on the packaging boxes (groups of 25 cells per box).

In order to compare batches B1 and B2, the respective result plots of a certain parameter are displayed in the same row: at the left hand side of the figure the parametric variation of batch B1 is displayed in blue; the right hand side shows the corresponding parametric variation of batch B2 in orange.

4.1.1. Variation of discharge capacities

Fig. 3 shows the parametric variations of the discharge capacities C_DCH_CC, C_DCH_CV and C_DCH_CCCV.

The discharge capacity obtained from the CC-phase C_DCH_CC shows similar absolute mean and standard deviation values for both batches B1 and B2. The respective relative variation is small: $\sigma/\mu|_{C_DCH_CC,B1} = 0.41\%$, $\sigma/\mu|_{C_DCH_CC,B2} = 0.48\%$. Neither group building nor any drift can be observed for C_DCH_CC.

The discharge capacity obtained from the CV-phase C_DCH_CV is slightly left skewed for both batches B1 and B2. The absolute mean and standard deviation values are similar for both batches. The respective relative variation is large: $\sigma/\mu|_{C_DCH_CV,B1} = 7.94\%$, $\sigma/\mu|_{C_DCH_CV,B2} = 8.65\%$. Neither group building nor any drift can be observed for C_DCH_CV.

The discharge capacity obtained from both discharge phases C_DCH_CCCV shows two peaks for both batches B1 and B2. The group building is also observed in the respective scatter plots. However, it cannot be attributed to the measure of 25 cells per box. The absolute mean and standard deviation values are again similar for both batches and the relative variation is small: $\sigma/\mu|_{C_DCH_CCCV,B1} = 0.23\%$, $\sigma/\mu|_{C_DCH_CCCV,B2} = 0.33\%$.

As the relative MU is very small for all capacities ($u(x)/\sigma_x < 2\%$, comp. Table 3), the determined variation results are not affected by the MU of the capacity values.

The relative variation σ_x/μ_x of both, C_DCH_CC and C_DCH_CCCV, is smaller than 0.5%. However, the variation of C_DCH_CV is much higher around 8%. The reason for the relative variation of C_DCH_CV being so much bigger compared to C_DCH_CC, is assumed to be due to the fact that the CV-phase has both, a smaller duration and a smaller Ah-throughput than the CC-phase. In consequence, a small change in the duration of the CV-phase has a much higher influence on the result than the same change in duration on the result of the CC-phase. The same holds true for a change in the Ah-throughput. The reason for the relative variation of C_DCH_CCCV being the smallest value of all capacities, is the decreased influence of the variation of the cell impedance. This is due to C_DCH_CC and C_DCH_CV not being independent from each other: if the cell has a higher cell impedance, the arising overpotentials are higher, the voltage stop criterion is reached earlier; in consequence the CC-phase is shorter. Moreover, reducing higher overpotentials takes more time; therefore the CV-phase is longer compared to a cell with a smaller cell impedance. The CC-phase dominates the C_DCH_CCCV value due to its longer duration and higher Ah-throughput. Hence, when summing up both capacities to C_DCH_CCCV, the influence of the variation of the cell impedance is decreased and furthermore, the relative variation of C_DCH_CCCV is even smaller than that of C_DCH_CC.

4.1.2. Variation of impedances

Fig. 4 compares the three DC impedances R_DC_10s, R_DC_30s and R_DC_60s. The data of all DC impedances is normally distributed for both batches B1 and B2. The respective scatter plots neither show group building nor drift along the sequential cell number. The absolute mean impedance values and standard deviations of the two batches are similar, the relative variation of B1 is slightly higher compared to B2 for all three DC impedances. For R_DC_10s: $\sigma/\mu|_{R_DC_10s,B1} = 1.43\%$, $\sigma/\mu|_{R_DC_10s,B2} = 1.23\%$ (comp. Fig. 4 first row). For R_DC_30s: $\sigma/\mu|_{R_DC_30s,B1} = 1.34\%$, $\sigma/\mu|_{R_DC_30s,B2} = 1.22\%$ (comp. Fig. 4 second row). For R_DC_60s: $\sigma/\mu|_{R_DC_60s,B1} = 1.31\%$, $\sigma/\mu|_{R_DC_60s,B2} = 1.23\%$ (comp. Fig. 4 third row).

Thus, when determining the DC impedance after a longer period of time ($t=30s$, $t=60s$), the relative variation decreases insignificantly for batch B1, and it stays fairly constant for batch B2. Hence, the relative variation of a DC impedance is not influenced significantly by the time period of its determination and it is moreover similar for the two examined batches.

The MUs of the DC impedances are in the range of 24–33% of the determined standard deviation (comp. Table 3). Thus, about one quarter to one third of the determined variations could be due to the MUs.

Fig. 5 compares the two ohmic AC impedances R_AC_1m0 and R_AC_1000Hz. When analysing the ohmic AC impedances R_AC_1m0 and R_AC_1000Hz, the first thing to notice is that batch B1 shows two peaks within its respective histogram distributions instead of only one. The data is normally distributed within the single separate peaks. However, regarding the complete data set of batch B1, the data is not normally distributed. For comparative reasons with the other parameters, a normal

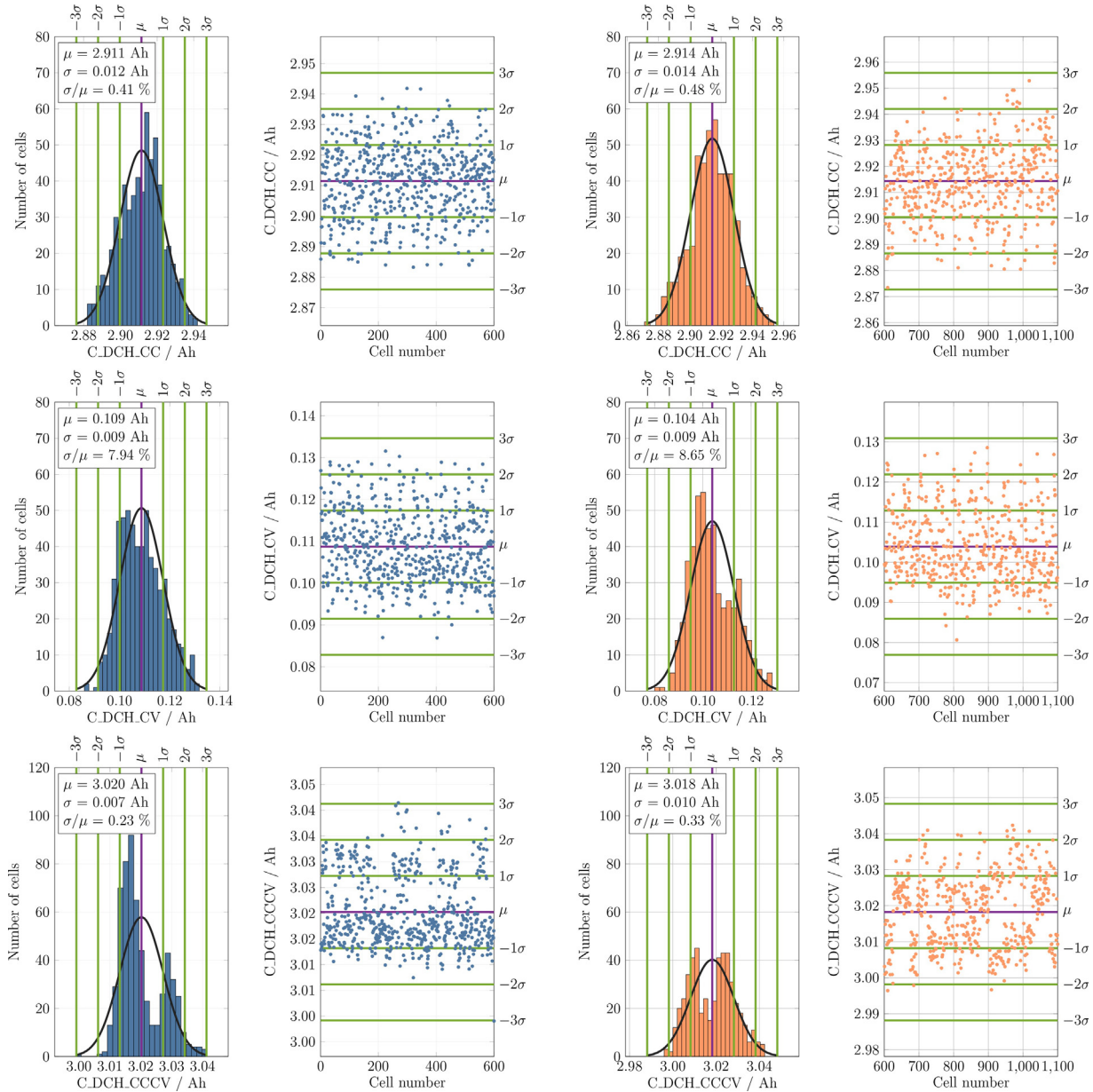


Fig. 3. Variations of the discharge capacity during only CC-phase (first row), only CV-phase (second row) and combined CCCV-phase (third row). On the left side batch B1 is displayed in blue, on the right side batch B2 is displayed in orange. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

distribution is nevertheless used for statistic analysis. The group building within batch B1 can also be observed within the scatter plots. Batch B2 shows a normal distribution for both ohmic AC impedances.

The absolute values of the two parameters are again similar for both batches, however the standard deviation for batch B1 is much higher than for B2 due to the distribution shape. Therefore, the respective relative variations of B1 are higher than those of B2. For R_{AC_Im0} : $\sigma/\mu|_{R_{AC_Im0},B1} = 1.82\%$, $\sigma/\mu|_{R_{AC_Im0},B2} = 0.74\%$. For R_{AC_1000Hz} : $\sigma/\mu|_{R_{AC_1000Hz},B1} = 1.78\%$, $\sigma/\mu|_{R_{AC_1000Hz},B2} = 0.71\%$. (comp. Fig. 5 first and second row).

For batch B1, the relative MUs of the ohmic AC impedances are about one third of the respective standard deviation ($u(x)/\sigma_x \approx 33\%$). For batch B2, it is much higher at about $u(x)/\sigma_x \approx 80\%$ (comp. Table 3). This is due to the fact, that the variation of B1 is much wider spread due to the observed group building. Hence, the

same MU has a larger influence on the less varying batch B2 than on B1.

An explanation for the observed two peaks within the histogram plots of the ohmic AC impedances of batch B1 could be a manufacturing adjustment during the production process. Such adjustments could be due to a change of material or due to a change in the process parameters. An example for a change of material could be a change of an electrode roll, a different current collector or a different electrolyte with a slightly different concentration. An example for a change in the process parameters could be a change in the welding parameters or tolerances concerning the tabs which connect jelly roll and cell can (i.e. length of the tabs, folding of the tabs during the insertion of the jelly roll into the can). As this clear group building within batch B1 cannot be observed for any other determined cell parameter but for the ohmic AC impedances, the latter reasons appear more realistic to

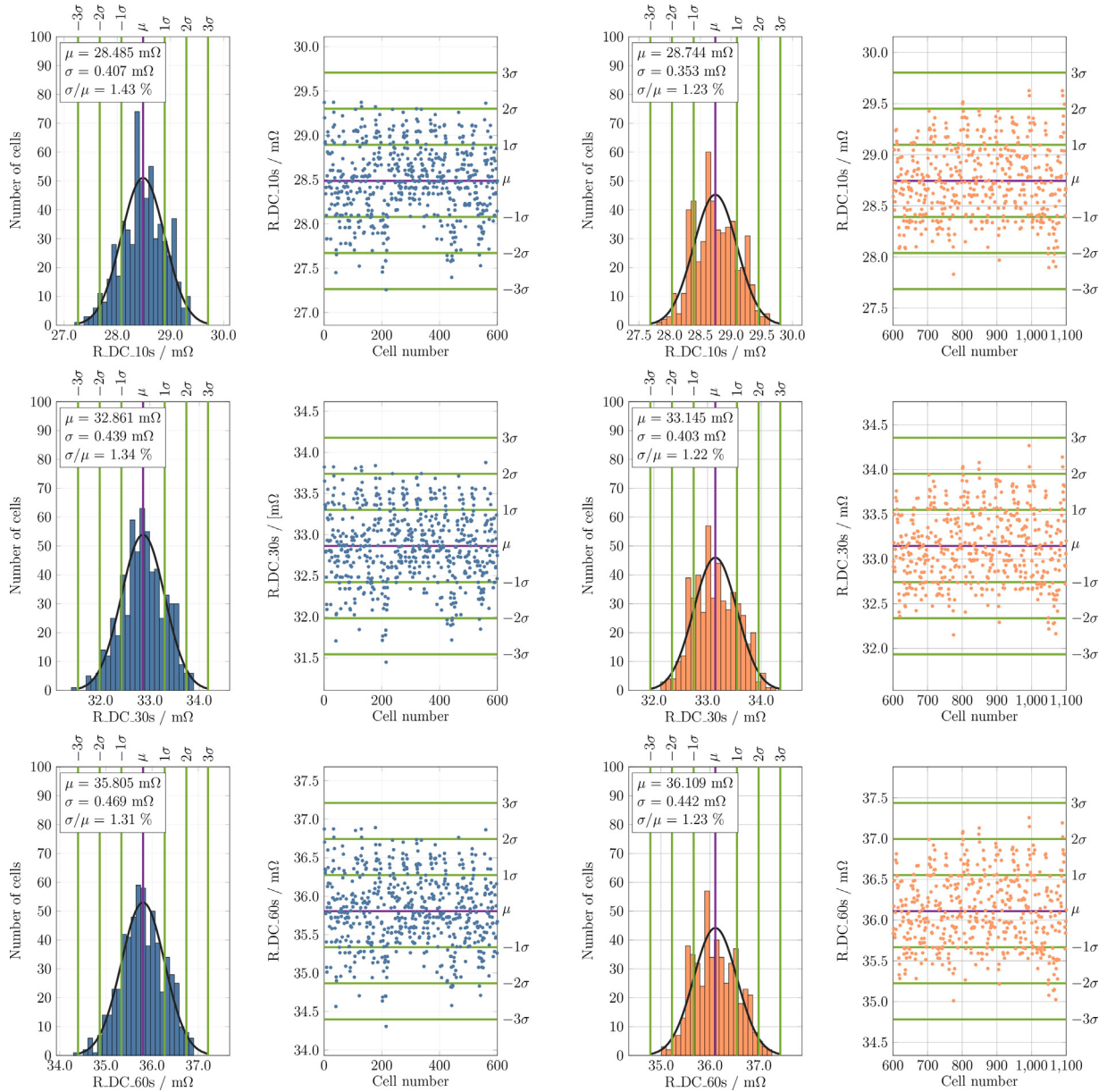


Fig. 4. Variations of the DC impedances after 10 s R_DC_10s (first row), 30 s R_DC_30s (second row) and 60 s R_DC_60s (third row). On the left side batch B1 is displayed in blue, on the right side batch B2 is displayed in orange. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the authors than a change in material. However, this is only an assumption as long as no additional information about the production process is provided by the manufacturer.

When comparing the two AC impedances within each single batch, the two AC impedances statistically show similar absolute and relative values. Hence, as there is no significant deviation between the two AC ohmic impedances, it is justifiable to use the R_AC_1000Hz impedance for experimental determination of the ohmic impedance of a cell. Within this work however, we will use the true ohmic impedance R_AC_Im0 for further analysis.

Fig. 6 compares the local extrema AC impedances R_AC_max and R_AC_min. They are both normally distributed for both batches. Neither group building nor drift can be observed within the scatter plots. The absolute mean values are slightly higher for batch B1, the absolute standard deviation values are similar for both batches regarding R_AC_max but higher for B2 regarding

R_AC_min. Therefore, the relative variation values are higher for B2 for both parameters. For R_AC_max: $\sigma/\mu|_{R_AC_max,B1} = 7.94\%$, $\sigma/\mu|_{R_AC_max,B2} = 8.37\%$ (comp. Fig. 6 first row). For R_AC_min: $\sigma/\mu|_{R_AC_min,B1} = 2.60\%$, $\sigma/\mu|_{R_AC_min,B2} = 3.08\%$ (comp. Fig. 6 second row).

Comparing the mean relative MU of both batches: it is about $u(x)/\sigma_x \approx 54\%$ for R_AC_max and higher at about $u(x)/\sigma_x \approx 88\%$ for R_AC_min (comp. Table 3). Hence, for the local maximum in the Nyquist plot, the variation could be 54% be due to the MU of the parameter, for the local minimum even by 88%.

The relative variation of R_AC_min is high around 2.8%, and even higher for R_AC_max around 8.2%. The reason for these high relative variation values could be the way of determining these AC parameters: the local maximum and minimum AC impedance values are determined by interpolating the real part values of the complex impedance from the Nyquist plot. As the complex

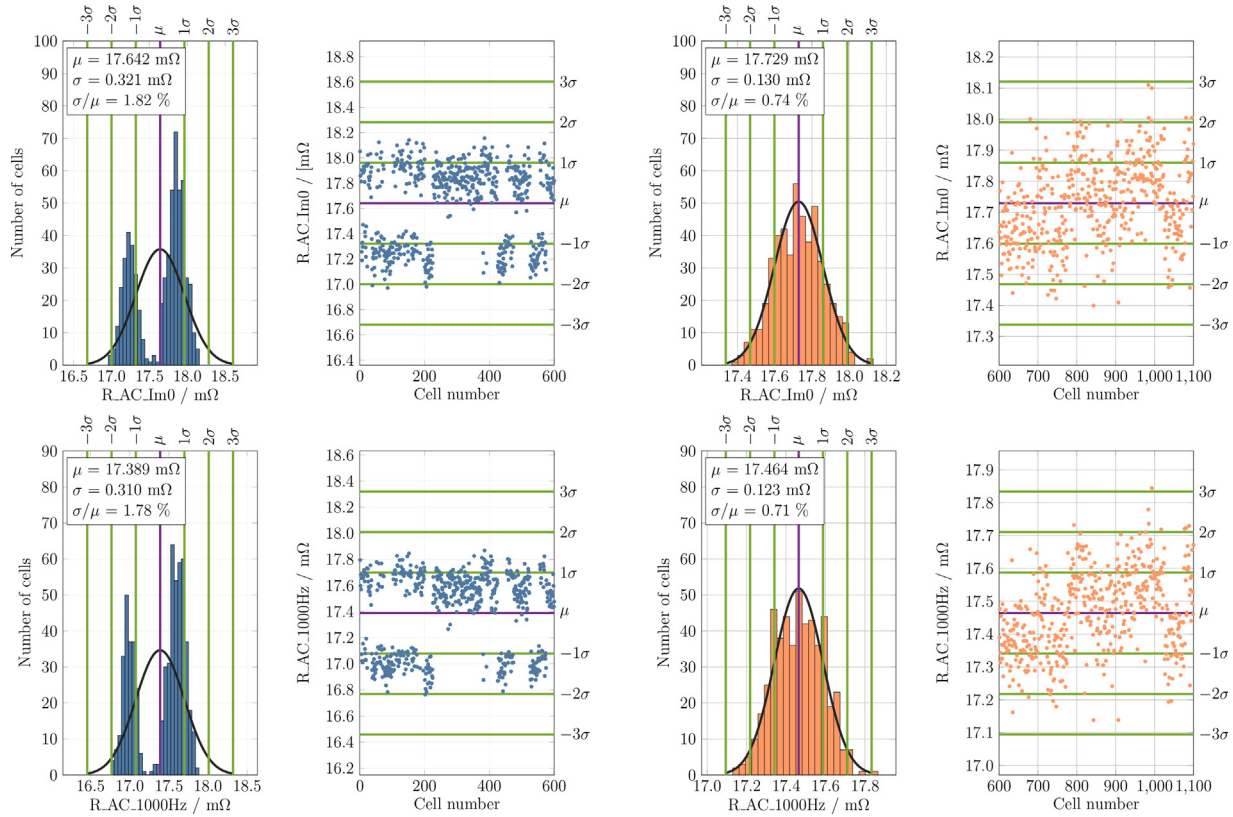


Fig. 5. Variations of the ohmic AC impedances R_{AC_Im0} (first row) and $R_{AC_1000\text{ Hz}}$ (second row). On the left side batch B1 is displayed in blue, on the right side batch B2 is displayed in orange. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

impedance has been determined at logarithmic spaced frequencies, only a limited number of values serve as input for the interpolation, especially within the lower frequency ranges. Hence, an oscillation of the originally measured values might not completely be smoothed by the interpolation and could thus result in a large variation of the determined values. Moreover, the plateau, where the local maximum is determined is quite flat and the local minimum is quite smooth within the Nyquist plot (comp. Fig. 2 right). This could also explain the high relative variations for the parameters. In order to discuss the influence of the frequency values on the variation of the AC impedances, the variation of the AC frequencies will be analysed in Section 4.1.3.

Comparing the absolute values of R_{DC_10s} and R_{AC_Im0} it can be noted that the mean value of R_{DC_10s} is much higher than that of R_{AC_Im0} . This is due to the fact that within a time-dependent determination of DC impedances the ohmic drop occurs within the first instant after a current pulse has been cut off. At a time $t = 10\text{ s}$ after the pulse not only the ohmic overpotential but also charge-transfer and diffusion overpotentials contribute to the measured impedance value. Hence, the ohmic DC impedance must be considered part of the R_{DC_10s} value, which is why this value deviates significantly from the ohmic AC impedance absolute values. Calculating the ohmic DC time constant by using the measured f_{AC_Im0} mean value considering both batches ($\mu|_{f_{AC_Im0,B1,B2}} \approx 761\text{ Hz}$), this would imply to determine an ohmic DC impedance after $t \approx 1.3\text{ ms}$. In fact, the R_{DC_1ms} value had been determined from the CU-measurements. However, this value showed an unrealistic huge relative variations (not depicted in this paper). This variation is attributed to the fact that 64 cells have been measured simultaneously by the CTS: a precise logging after such a small period of time seems to not have been possible for this number of cells. In other words, the large variation of R_{DC_1ms} is

interpreted to be due to measurement uncertainties and has therefore not been depicted in this work. Determining the DC impedance value after $t = 1\text{ s}$ (R_{DC_1s}) could possibly have resulted in a recordable value which might have been considered an approximate ohmic DC impedance. However, based on the investigated data, in order to analyse the cell-to-cell variation of the cell impedance it is necessary to determine an ohmic AC impedance as the DC impedances do not provide information about obvious group building within a batch of cells.

The following discussed values refer to the combined mean values of both batches. The R_{DC_10s} absolute mean value is about $\mu|_{R_{DC_10s}} \approx 28.5\text{ m}\Omega$, its corresponding frequency is $f|_{R_{DC_10s}} = 0.1\text{ Hz}$. This mean DC impedance value is already bigger than the sum of $R_{AC_Im0} + R_{AC_min} \approx 17.7\text{ m}\Omega + 5.5\text{ m}\Omega = 23.2\text{ m}\Omega$. Hence, the R_{DC_10s} impedance can be considered not only to include charge transfer processes but to additionally already include diffusion processes. The further DC impedances R_{DC_30s} and R_{DC_60s} even include more diffusion processes. The mean corresponding frequency to R_{AC_min} is $f_{AC_min} \approx 3.6\text{ Hz}$. A DC impedance, which was comparable to $R_{AC_Im0} + R_{AC_min}$ would thus have to be measured after $t \approx 0.3\text{ s}$.

4.1.3. Variation of AC frequencies

Fig. 7 compares the interpolated AC frequencies f_{AC_Im0} , f_{AC_max} and f_{AC_min} . As defined in Section 3.7, the AC impedances are determined by interpolating the measured frequency values. This gives fixed frequency values for which the respective impedance is found. Fig. 7 shows, how these frequencies vary for the two batches. f_{AC_Im0} does not show any drift or group building within the scatter plot. The relative variation of the parameter is in a similar range for the two batches, however, it is higher for batch B1: $\sigma/\mu|_{f_{AC_Im0,B1}} = 4.57\%$,

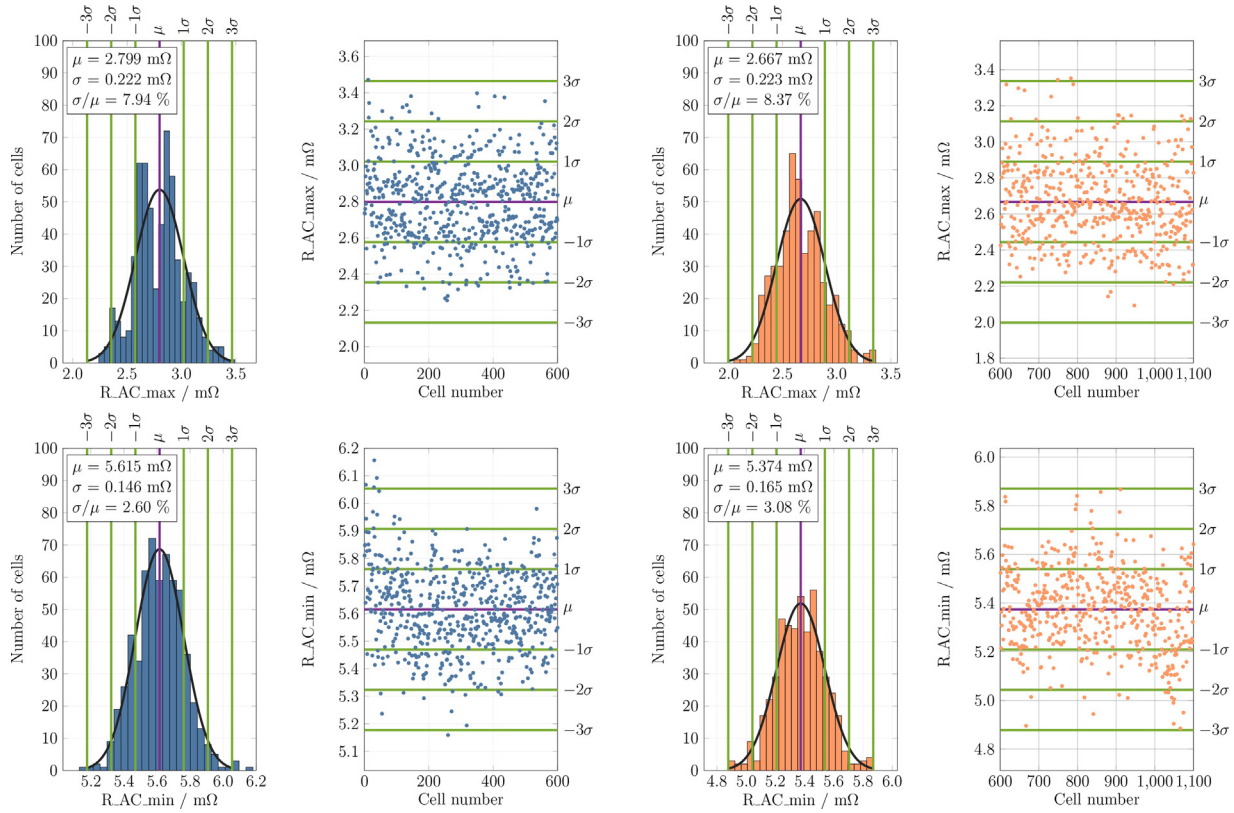


Fig. 6. Variations of the local maximum AC impedance R_{AC_max} (first row), and the local minimum AC impedance R_{AC_min} (second row). On the left side batch B1 is displayed in blue, on the right side batch B2 is displayed in orange. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$\sigma/\mu|_{f_{AC_Im0,B2}} = 3.79\%$ (comp. Fig. 7 first row). The local maximum and minimum values are found at fixed interpolated frequencies values. Therefore, the variation of these frequencies is not scattered but lined at the respective fixed frequency values. As the interpolation point density was not very high around these local extrema, the frequency range is big, which leads to a high relative variation. For f_{AC_max} : $\sigma/\mu|_{f_{AC_max,B1}} = 15.42\%$, $\sigma/\mu|_{f_{AC_max,B2}} = 16.33\%$ (comp. Fig. 7 second row). For f_{AC_min} : $\sigma/\mu|_{f_{AC_min,B1}} = 20.75\%$, $\sigma/\mu|_{f_{AC_min,B2}} = 22.59\%$ (comp. Fig. 7 second row).

This large variation in frequencies of the local extrema explains the determined big variation in impedance values of the local extrema (comp. Section 4.1.2). This implies, that the EIS-data is very sensitive to the chosen frequency density during an EIS-measurement. This especially applies to the low frequency range, within which the local maximum and minimum is determined. Choosing an increased number of low frequency points during an EIS-measurement means accepting a prolonged measurement time. However, it should be kept in mind that when aiming to fit an ECM by using these local maximum and minimum values, a preferably high accuracy of these characteristic data points is crucial for the quality of later modelling results. Another option would be to find the most suitable fixed frequencies for the characteristic data points – either by pre-measurements or by successive approximation. In any case, a good compromise should be found between measurement time and fitting accuracy.

4.1.4. Variation of mass and temperature

Fig. 8 shows the parametric variations of the mass and mean temperature during the CU-measurement T_{Mean} .

The mass of the cells is normally distributed for both batches. The absolute mean and standard deviation values are similar for both batches and the relative variation is small: $\sigma/\mu|_{Mass,B1} = 0.07\%$, $\sigma/\mu|_{Mass,B2} = 0.12\%$ (comp. Fig. 8 first row). Batch B2 shows a slight increase in mass with increasing cell number.

The corresponding relative MU is about $u(x)/\sigma_x < 1\%$ (comp. Table 3). Hence, the MU has no impact on the variation result.

The mean temperature during the CU-measurement T_{Mean} is normally distributed for both batches. The absolute mean and standard deviation values are similar for both batches and the relative variation is equal: $\sigma/\mu|_{T_{Mean,B1}} = 2.56\%$, $\sigma/\mu|_{T_{Mean,B2}} = 2.56\%$ (comp. Fig. 8 second row).

For T_{Mean} the relative MU is about $u(x)/\sigma_x \approx 16\%$ (comp. Table 3). This is not expected to significantly affect the variation results.

Moreover, no drift or group building can be observed within the scatter plots. This is a crucial finding as it implies three facts: first, the surrounding temperature within the climate chamber did not show a gradient, i.e. from bottom to top of the chamber, as no temperature drift within groups of 64 cells (maximum number of cells within the climate chamber) can be noticed. Second, the temperature was constant over the whole time of the CU-measurement, as no drift over the sequential cell number is apparent. Third, none of the terminal boards (4 cells per board) has been defective which could have led to an increased temperature of single cells in a regular cell number interval. Hence, the prior met assumptions regarding the temperature of the cells have been justified and errors within parametric variations due to the temperature can be excluded. It must be kept in mind though, that T_{Mean} is the mean temperature during the CU-measurement.

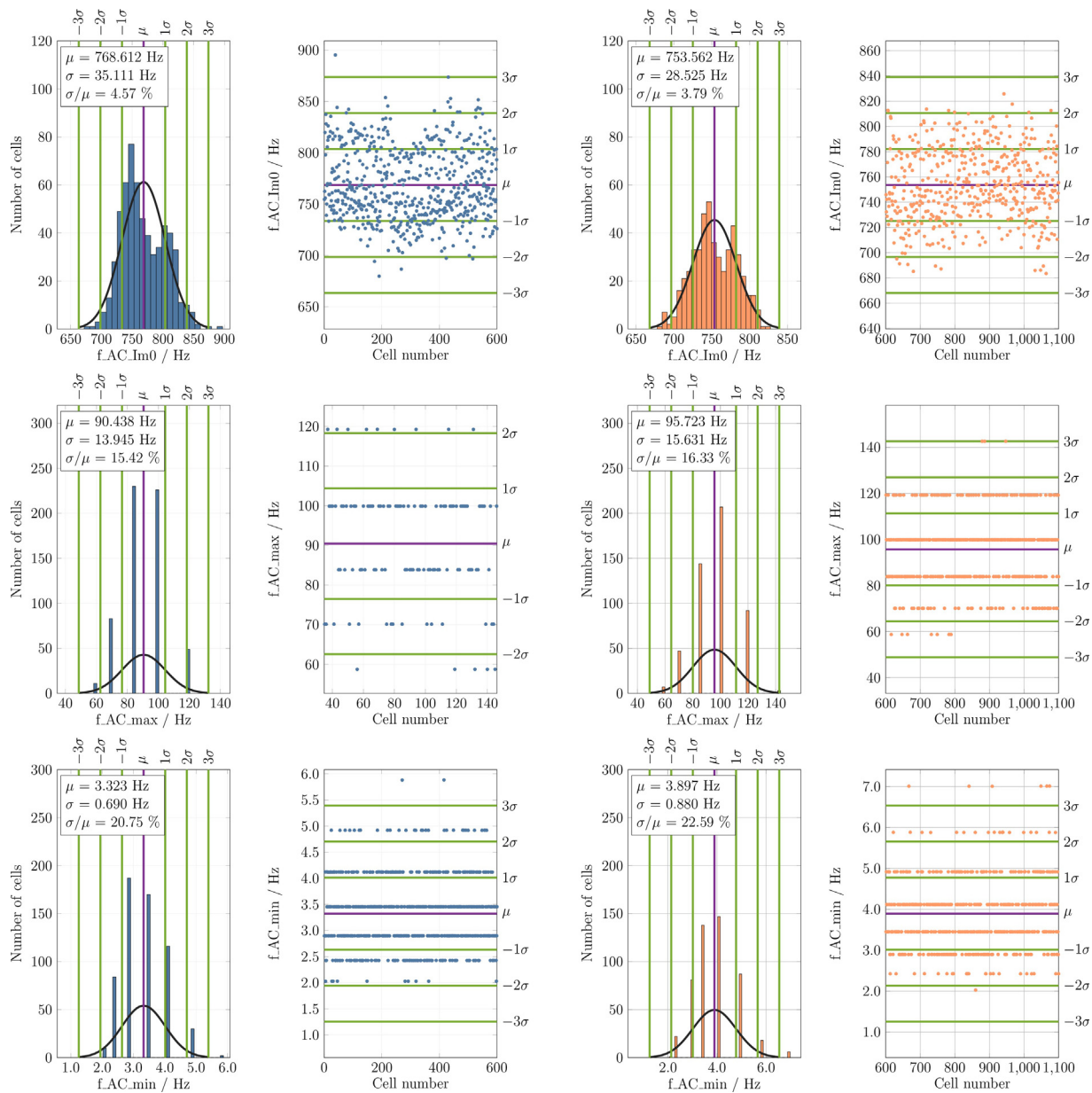


Fig. 7. Variations of the interpolated AC frequencies f_{AC_Im0} (first row), f_{AC_max} (second row) and f_{AC_min} (third row). On the left side batch B1 is displayed in blue, on the right side batch B2 is displayed in orange. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Hence, the significance of a correlation between this temperature parameter and the parameters determined from the EIS-measurement is limited.

4.1.5. Comparison of parametric variations

Table 3 summarizes the parametric variations of the investigated parameters and states their relative MUs.

C_{DCH_CC} and C_{DCH_CV} both contribute to the overall discharge capacity C_{DCH_CCCV} . For C_{DCH_CCCV} the influence of the variation of the cell impedance is decreased. Therefore, its relative variation is the smallest $\sigma/\mu \approx 0.28\%$ compared to the other capacities. C_{DCH_CC} shows only a slightly higher value $\sigma/\mu \approx 0.45\%$. C_{DCH_CV} seems to be prone to technical measurement related variations. The relation to cell internal parametric variation remains to be investigated, e.g. by simulation.

Regarding the investigated cell impedances, determining the ohmic cell impedance proves to be necessary in order to determine

a group building within a batch of cells, which has only been observed for the ohmic AC impedances. The DC impedances have not allowed for this classification. As the two ohmic AC impedances R_{AC_Im0} and R_{AC_1000Hz} show the same statistic behaviour, it is justifiable to use R_{AC_1000Hz} to determine the ohmic cell impedance during cell characterisation. R_{AC_1000Hz} can also be determined at $f = 1000$ Hz by using an impedance meter or battery tester.

Without considering batch B1 (which shows group building), the DC impedances show a constant relative variation $\sigma/\mu \approx 1.22\%$ independent of the chosen time interval, whereas the ohmic AC impedances show a relative variation $\sigma/\mu \approx 0.72\%$, which is smaller by 40%. The local maximum and minimum impedances show a higher relative variation (about 7 times and 2.5 times higher respectively). These strong variations are interpreted to be due to the strong variation of their corresponding frequencies. Hence, when aiming to use the local maximum and minimum AC

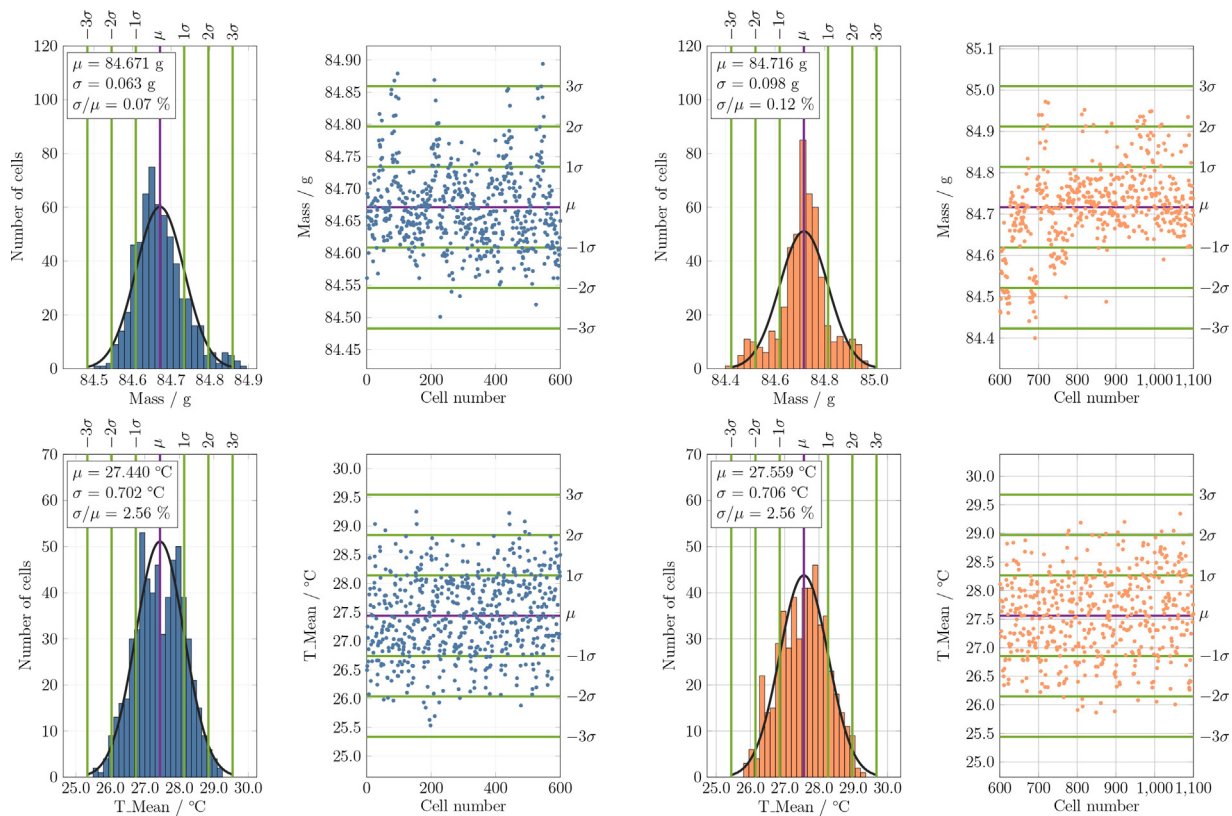


Fig. 8. Variations of the mass (first row) and mean temperature during the CU-measurement T_{Mean} (second row). On the left side batch B1 is displayed in blue, on the right side batch B2 is displayed in orange. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

impedances for ECM fitting purposes and reliable modelling results, the density of frequency points during the EIS-measurement should be chosen carefully, especially when a flat local maximum plateau and smooth local minimum within the Nyquist diagram are observed.

Cell-to-cell variations of capacity and impedance reported in literature are within the same range as our results (comp. Table 4). Based on our determined results for batch B2, the ratio between the respective relative variations of $C_{\text{DCH_CCCV}}$ and the ohmic impedance $R_{\text{AC_Im0}}$ equals $\sigma/\mu|_{C_{\text{DCH_CCCV}}}/\sigma/\mu|_{R_{\text{AC_Im0}}} \approx 1:2.2$. When considering $C_{\text{DCH_CC}}$, the ratio equals $\sigma/\mu|_{C_{\text{DCH_CC}}}/\sigma/\mu|_{R_{\text{AC_Im0}}} \approx 1:1.5$. These ratios are a little smaller than those reported in literature (comp. Table 4). However, it becomes obvious that when aiming to relate and compare cell-to-cell variations, it is crucial to clearly identify the type and way of determination of the respective parameters. Otherwise the values are difficult to compare and interpret.

4.2. Parametric correlations between cell parameters

In order to investigate the dependency between two determined cell parameters, their respective correlation graph is analysed. The x- and y-coordinates of each correlation graph are defined by the range of parametric variation of the two respective parameters. Illustrating the values for all cells of a batch within the correlation graph will form a scatter plot. A totally positive ($\rho_{xy} = 1$) (negative ($\rho_{xy} = -1$)) correlation of the two parameters will be shown by a straight line with a positive (negative) gradient. A totally missing correlation ($\rho_{xy} = 0$) is depicted by a non-biased scatter plot. In the lower right corner of each correlation graph, the correlation coefficient ρ (comp. Section 2.2) is given.

For being able to compare the correlation between different parameters, the single correlation graphs are arranged in a graphical correlation matrix: as the correlation coefficient between one parameter and itself always equals $\rho_{xx} = 1$ and the plot is therefore not of interest (straight line), the diagonal cells of

Table 4

Overview of experimental investigations on cell-to-cell variations of capacity and impedance of production fresh cells reported in literature compared to our work: besides basic cell parameters, the resulting relative variations of capacity and impedance as well as the ratio between the two values are listed.

Reference	Dubarry [21]	Campestrini [16]	Schuster [17]	Paul [4]	Our work
Number. of new cells	10	250	484	20000	1100 (B2: 500)
Chemistry	–	NCA	NMC	LFP	LFP
Nominal capacity	1.9 Ah	2.8 Ah	1.95 Ah	4.4 Ah	3.0 Ah
Cell ID	Commercial 18650	Panasonic NCR18650PD	Molicel IHR18650A	–	Sony US26650FTC1
$\sigma/\mu _{\text{capacity}}$	$C_{\text{DCH_CCCV}}$: 0.16%	$C_{\text{DCH_CCCV}}$: 0.16%	$C_{\text{DCH_CCCV}}$: 0.8%	C_{DCH} : 1.3%	$C_{\text{DCH_CCCV}}$ (B2): 0.33%
$\sigma/\mu _{\text{impedance}}$	R_{DC} : 5.66%	$R_{\text{AC_1000 Hz}}$: 0.72%	$R_{\text{AC_Im0}}$: 1.94%	R_{DC} : 5.8%	$R_{\text{AC_Im0}}$ (B2): 0.73%
$\frac{\sigma/\mu _{\text{capacity}}}{\sigma/\mu _{\text{impedance}}}$	1:35.4	1:4.5	1:2.4	1:4.5	1:2.2

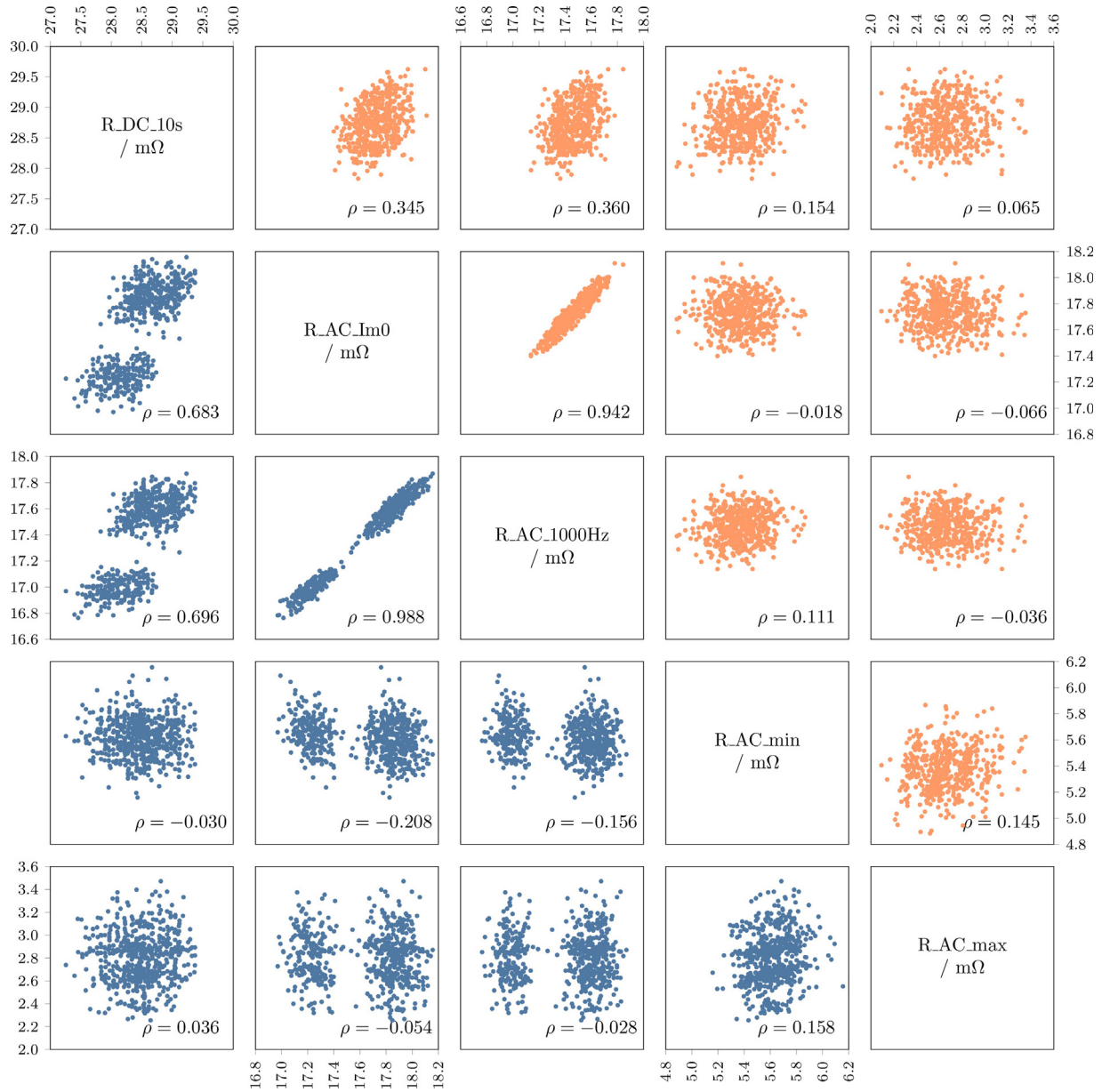


Fig. 9. Correlations between the following cell parameters: R_DC_10s, R_AC_Im0, R_AC_1000Hz, R_AC_min and R_AC_max. The blue points represent batch B1, the orange points batch B2. The correlation coefficient values range from $\rho \in [-1, 1]$ and are given in the lower right corner of each correlation plot. Their corresponding correlation classes are defined in Section 2.2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the graphical correlation matrix are used to state the name and unit of the respective parameter. Hence, the matrix graph is divided into two triangular matrices by this diagonal line. The lower triangular matrix represents the parametric correlation data of batch B1 (blue), the upper triangular matrix that of batch B2 (orange). Each graphical matrix cell displays the single correlation graph of those two parameters which are given in the respective row and column. The given parametric ranges relate to all correlation graphs within the respective row or column.

Fig. 9 shows the correlation between different impedances. Fig. 10 shows the correlation between selected impedances, discharge capacities, mass and temperature.

Table 5 summarizes the obtained correlation coefficients for all 15 parameters. The structure of the table is based on the structure of the graphical correlation matrices within the correlation figures.

4.2.1. Correlation between impedances

The three determined DC impedances R_DC_10s, R_DC_30s and R_DC_60s show a strong correlation between each other: $\rho_{xy} \in [0.984, 0.996]$ for both batches (comp. Table 5). Therefore, only R_DC_10s will be used for the following graphical analysis of the parametric correlations.

Fig. 9 shows the correlation between the following impedance values: R_DC_10s, R_AC_Im0, R_AC_1000Hz, R_AC_min and R_AC_max.

R_AC_Im0 and R_AC_1000Hz show a strong correlation for both batches ($\rho_{xy,B1} = 0.988$, $\rho_{xy,B2} = 0.942$). This is a crucial finding as it justifies the assumption that measuring the R_AC_1000Hz impedance during cell characterization does adequately represent the ohmic impedance of the cell.

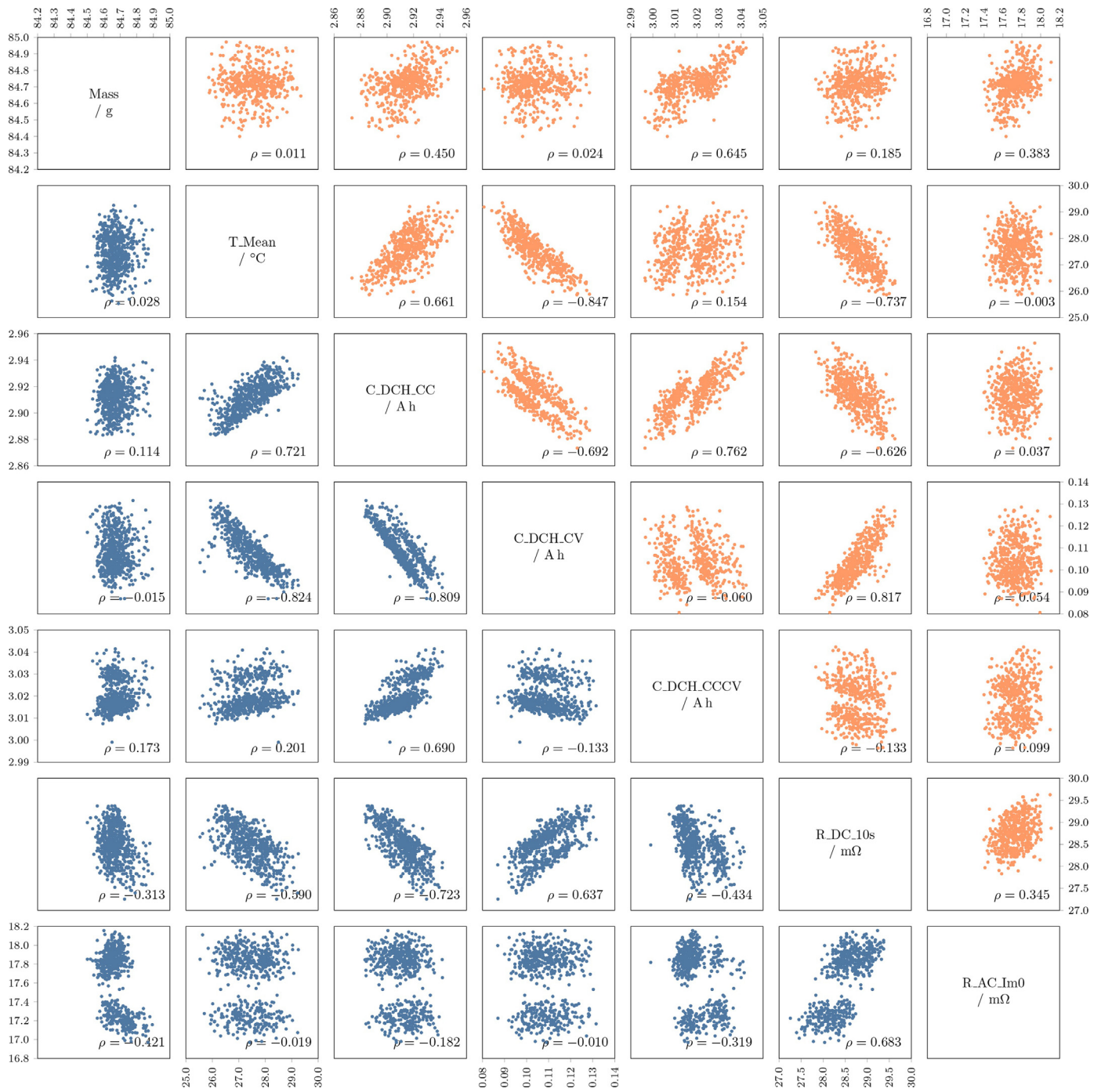


Fig. 10. Correlations between the following cell parameters: Mass, T_Mean, C_DCH_CC, C_DCH_CV, C_DCH_CCCV, R_DC_10s and R_AC_Im0. The blue points represent batch B1, the orange points batch B2. The correlation coefficient values range from $\rho \in [-1, 1]$ and are given in the lower right corner of each correlation plot. Their corresponding correlation classes are defined in Section 2.2. Note that T_Mean is the mean temperature during the CU-measurement. A correlation of T_Mean with AC values is therefore not significant.

For batch B1 two separate scatter plots can be observed within those correlation graphs, which are dependent on one of the ohmic AC impedances R_AC_Im0 or R_AC_1000Hz. This data separation within the scatter plot implies a group building within batch B1. The remaining correlation graphs of B1 do not show this group building. The impedances R_DC_10s, R_AC_min and R_AC_max do not represent pure ohmic impedances of the cell but are associated with charge transfer and diffusion processes. Hence, the partially observed group building seems to relate to a change in the pure ohmic impedance during the production process of batch B1. Thus,

batch B2 seems more suitable for further discussions of the correlation between parameters.

Regarding batch B1, R_DC_10s shows a moderate but nearly strong correlation with both, R_AC_Im0 ($\rho_{xy,B1} = 0.683$) and R_AC_1000Hz ($\rho_{xy,B1} = 0.696$). However, for B2 these correlations are only moderate and rather weak for both, R_AC_Im0 ($\rho_{xy,B2} = 0.345$) and R_AC_1000Hz ($\rho_{xy,B2} = 0.360$). This moderate correlation between the R_DC_10s and the ohmic AC impedances occurs due to the fact that the pure ohmic impedance of the cell contributes to the investigated DC impedance. However, the

Table 5

Experimentally determined parametric correlation coefficients between all 15 parameters: The lower triangular matrix gives the correlation coefficients of batch B1 in blue, the upper triangular matrix gives the correlation coefficients of batch B2 in orange. The definition of the given parameters is shown in Fig. 2. The correlation coefficient values range from $\rho \in [-1, 1]$. Their corresponding correlation classes are defined in Section 2.2. The boxes A, B, C, and D group the data from which four major implications are extracted for the development and modelling of battery modules (comp. Section 4.3).

Mass	0.011	0.450	0.024	0.645	0.185	0.180	0.182	0.383	0.400	-0.024	-0.014	0.007	-0.004	-0.004
0.028	T_Mean	0.661	-0.847	0.154	-0.737	-0.762	-0.772	-0.003	-0.017	-0.102	-0.031	-0.055	-0.102	0.017
0.114	0.721	C_DCH_CC	-0.692	0.762	-0.626	-0.643	-0.649	0.037	0.045	-0.143	-0.052	-0.047	-0.025	-0.021
-0.015	-0.824	-0.809	C_DCH_CV	-0.060	0.817	0.848	0.860	0.054	0.030	0.013	0.015	-0.088	-0.027	-0.019
0.173	0.201	0.690	-0.133	C_DCH_CCCV	-0.133	-0.128	-0.126	0.099	0.089	-0.187	-0.059	-0.144	-0.058	-0.046
-0.313	-0.590	-0.723	0.637	-0.434	R_DC_10s	0.992	0.990	0.345	0.360	0.154	0.065	0.051	-0.007	-0.063
-0.277	-0.641	-0.758	0.698	-0.421	0.992	R_DC_30s	0.996	0.325	0.337	0.150	0.067	0.042	-0.013	-0.063
-0.254	-0.669	-0.779	0.730	-0.415	0.984	0.995	R_DC_60s	0.314	0.324	0.146	0.068	0.034	-0.016	-0.062
-0.421	-0.019	-0.182	-0.010	-0.319	0.683	0.623	0.584	R_AC_Im0	0.942	-0.018	-0.066	-0.234	0.032	-0.084
-0.415	-0.034	-0.192	-0.009	-0.335	0.696	0.637	0.598	0.988	R_AC_1000Hz	0.111	-0.036	0.057	0.081	-0.096
0.033	-0.074	-0.114	0.048	-0.134	0.036	0.044	0.048	-0.054	-0.028	R_AC_max	0.145	0.143	-0.940	-0.026
0.048	-0.102	-0.151	0.023	-0.227	-0.030	-0.012	-0.006	-0.208	-0.156	0.158	R_AC_min	0.466	0.067	-0.767
0.163	-0.099	-0.040	-0.008	-0.077	-0.123	-0.106	-0.098	-0.373	-0.244	0.445	0.182	f_AC_Im0	0.127	-0.035
0.017	0.035	0.043	-0.047	0.016	-0.057	-0.058	-0.058	-0.054	-0.050	-0.943	0.046	0.058	f_AC_max	-0.003
0.012	0.002	0.004	0.031	0.044	0.022	0.017	0.018	0.026	0.012	0.043	-0.758	-0.108	-0.076	f_AC_min

change within the correlation classes when comparing batch B1 and B2, implies that the moderate but nearly strong correlation observed for B1 results from the group building within that batch.

Within all other correlation graphs the correlation is weak ($|\rho_{xy}| < 0.3$). In detail, this implies that the parameters R_{AC_min} and R_{AC_max} are neither dependent on each other nor dependent on R_{DC_10s} , R_{AC_Im0} or R_{AC_1000Hz} for neither batch B1 nor B2.

Hence, only the R_{DC_10s} and R_{AC_Im0} impedance will be used for the following investigation of correlation between impedances, discharge capacities, mass and temperature of the cells.

4.2.2. Correlation between mass, temperature, discharge capacities and specific impedances

Fig. 10 shows the correlation between the impedances R_{DC_10s} and R_{AC_Im0} , the discharge capacities C_{DCH_CC} , C_{DCH_CV} and C_{DCH_CCCV} , the mass and the mean temperature during CU-measurement T_{Mean} .

The discharge capacities show differently strong correlations between each other: The correlation between C_{DCH_CV} and C_{DCH_CCCV} is weak. The correlation between C_{DCH_CC} and C_{DCH_CV} is moderate strong to strong and negative ($\rho_{xy, B1} = -0.809$, $\rho_{xy, B2} = -0.692$), whereas the correlation between C_{DCH_CC} and C_{DCH_CCCV} is moderate strong and positive ($\rho_{xy, B1} = 0.690$, $\rho_{xy, B2} = 0.762$). This finding indicates that when a cell has been charged more (compared to the other cells) during the CC-phase, it is charged less (compared to the other cells) during the CV-phase and vice versa.

The correlation between the DC impedance R_{DC_10s} and the single phase discharge capacities is moderate strong to strong and negative for C_{DCH_CC} ($\rho_{xy, B1} = -0.723$, $\rho_{xy, B2} = -0.626$) and

positive for C_{DCH_CV} ($\rho_{xy, B1} = 0.637$, $\rho_{xy, B2} = 0.817$). This means, that a high CC discharge capacity correlates to a small DC impedance, whereas a high CV discharge capacity correlates to a high DC impedance. These results support the explanations met in Section 4.1: a higher cell impedance leads to higher overpotentials during the CC-phase which then take longer to be reduced during the CV-phase. For C_{DCH_CCCV} the influence of variation of the cell impedance is decreased by summing up C_{DCH_CC} and C_{DCH_CV} resulting in a weak to moderate weak and negative correlation between C_{DCH_CCCV} and R_{DC_10s} . Hence, the variation of R_{DC_10s} allows no conclusion about the variation of C_{DCH_CCCV} .

The correlation between the ohmic AC impedance R_{AC_Im0} and the discharge capacities is weak. This is due to the fact that during the discharge of a cell mainly those overpotentials are relevant, which are due to processes occurring at low frequencies. Hence, in order to make statements about the variation of both, the cell impedance and the cell capacity, both parameters need to be investigated.

The temperature T_{Mean} shows the same correlation behaviour for both batches: Between T_{Mean} and C_{DCH_CC} a moderate strong to strong positive correlation ($\rho_{xy, B1} = 0.721$, $\rho_{xy, B2} = 0.661$) is observed. Between T_{Mean} and C_{DCH_CV} a strong negative correlation ($\rho_{xy, B1} = -0.824$, $\rho_{xy, B2} = -0.847$) can be seen. Between T_{Mean} and R_{DC_10s} a moderate strong to strong negative correlation ($\rho_{xy, B1} = -0.590$, $\rho_{xy, B2} = -0.737$) is observed. A high impedance value leads to a shorter CC-phase which again leads to a smaller mean temperature value during the whole CU-measurement. The strong temperature dependency observed for the single phase discharge capacities is compensated when combining both capacities to C_{DCH_CCCV} ($\rho_{xy, B1} = 0.201$, $\rho_{xy, B2} = 0.154$). Hence, using this discharge capacity during cell characterisation is

beneficial, as the temperature dependency resulting from the correlation with the cell impedance is reduced. The significance of any correlation between T_{Mean} and the ohmic AC impedance $R_{\text{AC_Im0}}$ is limited as T_{Mean} is the mean temperature determined from the CU-measurement and not from the EIS-measurement.

The mass does not show any correlation with the other investigated parameters, except for a moderate correlation with $C_{\text{DCH_CCCV}}$ and only for B2 ($\rho_{xy,B2} = 0.645$). Such a correlation could be explained by a variation in cell internal parameters such as the volume fraction of active material. Unless corresponding information by the manufacturer can be accessed, this dependency between internal and external cell parameters can be investigated by physicochemical simulation.

Table 5 shows the obtained correlation coefficients for all 15 parameters. Analysing the dependency of the local maximum and minimum AC impedance on their corresponding frequency reveals that their respective correlation is very strong and negative: The correlation between $R_{\text{AC_max}}$ and $f_{\text{AC_max}}$: $\rho_{xy,B1} = -0.943$, $\rho_{xy,B2} = -0.940$. The correlation between $R_{\text{AC_min}}$ and $f_{\text{AC_min}}$: $\rho_{xy,B1} = -0.758$, $\rho_{xy,B2} = -0.767$. This finding confirms the assumption that it is crucial to carefully choose the frequencies and density of frequency points during an EIS-measurement in order to allow for accurate ECM fitting results.

4.3. Implications for developing and modelling battery modules

From the results, four major implications for the development and modelling of battery modules as well as for the quality control during cell production can be extracted. They are numbered according to the boxes which group the data in Table 5.

- A. For modelling purposes, each of the DC impedances can be used, as $R_{\text{DC_10s}}$, $R_{\text{DC_30s}}$ and $R_{\text{DC_60s}}$ all show the same statistic behaviour. Their correlation is very strong. Hence, two of the three DC impedances can be omitted during characterisation. However, the DC impedances fail to depict the group building which is observed for the ohmic AC impedances within a batch.
- B. To make a statement about the variation of the cell impedance, it is therefore necessary to determine the ohmic cell impedance. As the ohmic AC impedances $R_{\text{AC_Im0}}$ and $R_{\text{AC_1000Hz}}$ both show the same statistic behaviour, also a faster measurement technique using an impedance meter or battery tester, which determines the ohmic impedance at a constant frequency $f = 1000 \text{ Hz}$, can be considered. In this case, an EIS-measurement is not necessarily required and the four AC impedances could be omitted during cell characterisation.
- C. To make a statement about the variation of the cell capacity, $C_{\text{DCH_CCCV}}$ should be determined, as this is the capacity for which the influence of the variation of both, the cell impedance and the temperature is decreased. As this capacity calculates from the two other capacities $C_{\text{DCH_CC}}$ and $C_{\text{DCH_CV}}$, none of the determined capacities can be omitted during characterisation. Nevertheless, the measurement time does not increase. Note, that the variation of the cell impedance allows no conclusion about the variation of the cell capacity.
- D. When aiming to use characteristic points within the Nyquist plot (i.e. the local maximum or minimum) determined from an EIS-measurement for ECM fitting purposes, the frequencies should be chosen carefully. $R_{\text{AC_max}}$ and $R_{\text{AC_min}}$ have been found to strongly depend on their respective frequencies $f_{\text{AC_max}}$ and $f_{\text{AC_min}}$. Especially if the local maximum plateau within the Nyquist diagram is flat and the local minimum is smooth, the impedances should be determined at either fixed frequencies or the interpolated frequencies should

be based on a high frequency density within that characteristic range. Hence, a pre-measurement is sensible in order to find the most promising fixed frequency or frequency density for each characteristic point and to thus achieve a good compromise between a prolonged measurement time and a high fitting accuracy.

Based on the analyses of the cell-to-cell variations and correlations of the investigated cell, two indices are at least necessary to characterise the parametric variation among cells: The most effective and time efficient parameter to depict the impedance variation is the $R_{\text{AC_1000Hz}}$ impedance. To identify the variation of capacity the parameter $C_{\text{DCH_CCCV}}$ should be determined.

5. Conclusion

We have experimentally investigated the parametric cell-to-cell variation of 15 different cell parameters and the correlation between them. The work is based on 1100 commercial production fresh LiFePO_4 -graphite cells being part of two batches (B1: cell 1–600, B2: cell 601–1100). This presented combination is unique in literature: only little publications base their statistic analysis on more than 100 cells and often the parametric variation is only determined for a limited number of parameters.

The investigated cell shows small cell-to-cell parametric variations compared to other cells [4,17]. For most parameters the respective variations are similar for the two investigated batches.

The relative variation σ/μ of discharge capacity obtained during CC-phase is small (about 0.45%) and even smaller for $C_{\text{DCH_CCCV}}$ (about 0.28%) with the influence of the variation of the cell impedance being reduced. In comparison, the relative variation of the ohmic AC impedance is still small but a little larger (about 0.72%, when considering batch B2 which shows no group building within the batch). This corresponds to variation ratios between cell capacity and ohmic impedance of 1:1.5 (CC) and 1:2.2 (CCCV) respectively. Due to the strong correlation between the two ohmic AC impedances $R_{\text{AC_1000Hz}}$ and $R_{\text{AC_Im0}}$, it is justifiable to use the $R_{\text{AC_1000Hz}}$ impedance during cell characterisation. The DC impedances all show the same statistic behaviour (relative variation about 1.3%). A high CC discharge capacity correlates to a small DC impedance, whereas a high CV discharge capacity correlates to a high DC impedance. The DC impedances do not show the group building which can be observed for the ohmic AC impedances of batch B1. Hence, determining the ohmic cell impedance proves to be necessary during cell characterisation when aiming to determine the cell-to-cell-variation of the cell impedance.

For fitting purposes of ECMs, frequently used parameters are the ohmic AC impedance and the impedances of the local extrema within the Nyquist diagram $R_{\text{AC_max}}$ and $R_{\text{AC_min}}$. The relative variation of the latter two parameters (about 8.1% and 2.8%) is found to be dependent on their respective frequency, which varies strongly (about 15.9% and 21.6%). Hence, a good compromise needs to be found between the number of low frequency points during an EIS-measurement resulting in a prolonged measurement time and the quality of fitting.

An important result from investigating the correlation between cell parameters, is that the cell impedance influences both, the discharge capacities and the mean temperature of the cell. A higher cell impedance leads to higher overpotentials during the discharge of the cell – therefore, the CC-phase is shorter. Reducing these overpotentials takes longer which increases the duration of the CV-phase. Hence, as the temperature mainly increases during the CC-phase, also the mean temperature during the CU-measurement is

lower for cells with a higher impedance. For C_DCH_CCCV the influence of the variation of the cell impedance and therefore also of the temperature is reduced.

From these results, four major implications for the development and modelling of battery modules as well as for the quality control during cell production have been extracted. Our experimentally determined parametric variation values and drawn conclusions are valuable for model fittings and battery pack analyses which have up to now been based on assumptions about cell-to-cell variations. Note, that the investigated cells have been production fresh cells. Hence, the reported cell-to-cell variations and correlations refer to manufacturing tolerances and the operational behaviour of fresh cells and cannot simply be adopted for aged cells.

In order to identify the most prominent cell internal parameters influencing the measured parametric variations, physicochemical modelling of the cell is to be performed. Based on the results, the inhomogeneity within battery modules due to different cell-to-cell variations will be investigated. The variation results of capacity and impedance presented in this paper can serve as input parameters for the model. By modelling different module topologies and operation scenarios, the influence of cell-to-cell variations on connected cells can be analysed in detail. This will allow us to develop superior matching strategies for battery modules considering both, the variation of cell capacity and impedance.

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